Memory Is the Defence: Physics-Gated Continuum-Memory Receivers for Poisoning-Resilient Test-Time Learning

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***Abstract*—A neural receiver that learns at test time trains on pilots an adversary can write to. This paper argues that it should consolidate into persistent memory only what a label-free physical check verifies against that memory. Citadel-Rx applies this to DeepSIC with a fast level fitted per block and two slow levels updated by a gated, clipped average of faster states; the clip bounds how far any attacker moves the slowest level per consolidation. With a single memory, a cyclic redundancy check (CRC) test with rollback fails, as do the gates of Citadel-Rx: their only reference, linear minimum mean-square error (LMMSE) detection on the poisoned pilots, degrades too. On ten held-out seeds of the stationary benchmark channel, white-box poisoning (A1) at 0 dB signal-to-interference ratio raises the symbol error rate (SER) of Citadel-Rx by $+0.0188$, against $+0.768$ for tuned online DeepSIC; the gates reject 0.993 of the poisoned and 0.001 of the clean blocks, 0.180 after a legitimate channel redraw (SER 0.251, online DeepSIC 0.224). The adapted attacks tested, all at full budget and rejected on nearly every block, gave an excess not distinguishable from plain A1 with five seeds; for plain A1 the ordering holds with a second backbone and 8×32 arrays. The worst attack found without a channel change, A4 (pilots and data, 20 gradient steps), gives $+0.0478$; poisoning during one also raises the excess. Citadel-Rx fails at 16-QAM and in a coded 5G link over 3GPP urban-micro channels, where its network is weaker than LMMSE on clean blocks.**

***Index Terms*—Deep receivers, test-time learning, adversarial pilot poisoning, continual learning, physics-informed gating, MIMO detection, physical-layer security.**

I. Introduction

DEEP receivers replace or augment the model-based blocks of a receiver with neural networks trained on data [1]–[4]. Since a network trained in simulation meets a different channel in the field, the strongest deep receivers adapt at test time: online DeepSIC [5] re-trains its per-user networks from block to block, and Meta-DeepSIC [6] meta-learns an initialisation from past blocks so that the per-block fit converges faster. In [6] the training data of a block are its decoded codewords, re-encoded and used as labels when decoding succeeds, and no new pilots are sent; in the pilot-trained form attacked in [7] and studied here, the adaptation data are the only labelled symbols the receiver sees: a few hundred pilot bits per user per block.

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Code, configuration files, every run record, the statistics and the scripts that regenerate every table, figure and number of this paper are available at https://github.com/abraheemrashid99-sys/Citradel-Rx; three commands re-run the two campaigns and one (`make paper`) rebuilds every number, table and figure and the paper (Section V).

An adversary within range of the base station can add a perturbation to the received pilot part of a block. Wu *et al.* [7] showed that a perturbation of the pilots, crafted on a surrogate receiver without knowledge of the target, transfers to online DeepSIC, to Meta-DeepSIC and to two other online detectors and impairs their adaptation. In the runs reported here (Section VI-A), a white-box perturbation shaped through an unrolled copy of the receiver's adaptation step, at a signal-to-interference ratio of 0 dB, raises the symbol error rate (SER) of online DeepSIC by $+0.768$ over its clean run, and Meta-DeepSIC, which keeps the poisoned pilots in its support set, still carries an excess after the attack stops. The receiver cannot verify its adaptation data, because the pilots are the labels. It can verify physical consistency: whether its decisions fit a linear channel model, whether the cyclic redundancy check (CRC) of the decoded data passes, and whether the estimated channel changed as a fading channel does from one block to the next. The harm of poisoning lies less in the attacked block than in what the receiver keeps.

This paper argues for one principle: *a test-time receiver resists pilot poisoning when it consolidates into persistent memory only what an independent, label-free physical check verifies, and when that check judges every update against the protected memory.* It is a design condition supported by an ablation, not a guarantee: a receiver that meets it still fails where its network is weaker than the linear minimum mean-square error (LMMSE) detector on clean blocks, where the attacker also degrades the references through the data, and during a legitimate channel change (Section VII). The two parts depend on each other. The simplest receiver with a check and a memory, online DeepSIC with a CRC test and rollback to its last accepted weights, fails. Its only reference is LMMSE on the same poisoned pilots, which the attack degrades together with the update. The first poisoned update therefore passes the relative test and becomes the state the receiver rolls back to, and every later update starts from it (Section VI-B). The receiver proposed here, Citadel-Rx, fails in the same way when it is given its gates but only a single memory (Section VI-G). Citadel-Rx keeps memory at three rates, a fast level that fits every block and two slow levels that change only through a gated, clipped average of the faster states, and its gates compare the fast level with the better of LMMSE on the pilots of the block and a detector built on the slow memory (the anchor detector), which those pilots do not reach. Multi-rate memory is one realisation of the principle; the ablation shows which of its parts are necessary.

TABLE I
POSITIONING. ATTACK SURFACE: WHAT THE ADVERSARY WRITES TO. ONLINE DEEPSIC AND META-DEEPSIC ARE LISTED IN THE PILOT-TRAINED FORM ATTACKED IN [7].

Class	Adapts	Memory	Attack surface / defence
LMMSE	no	–	pilot estimate / –
Pilot-spoofing detection [8], [9]	no	–	pilots / detection only
Online DeepSIC [5]	every block	last block	pilots / –
Meta-DeepSIC [6]	every block	pilot buffer	pilots, persistent / –
Code-verified online training [10], [11]	on decoded data	last block	labels / code check, no persistent gate
CRC-Rollback (Section VI-B)	every block	last accepted	pilots / CRC gate, rollback
Filtering and restoring TTA [12]–[15]	every batch	weights, anchor	batch / entropy filter, restore
Robust TTA [16], [17]	every batch	batch statistics	batch / robust statistics
Anchored fine-tuning [18]–[20]	every step	prior or averaged weights	data / pull to a prior or a teacher
Nested, continuum memory [21], [22]	every token	one or several rates	input / –
Citadel-Rx (this paper)	every block	three rates, gated	pilots / physics gates, protected fallback, bounded retention

Table I places the design among the work it draws on. Pilot spoofing is an established physical-layer threat [23], [24], with detectors that flag it [8], [9], but that work has no learning receiver to protect. Decision-directed training on decoded data verified by the code is the closest precedent for using the code check to decide what a receiver learns from [10], [11], and Meta-DeepSIC was proposed with it [6]; it verifies labels, not what enters a persistent memory. In deep receivers, drift detection decides when, and which module, to re-train [25], and Bayesian tracking replaces the multi-step fit of a block by a single-step update of a posterior over the weights [26]; both address channel variation, not an adversary. In vision, test-time adaptation (TTA) filters, anchors or restores what enters the adapted weights [12]–[15], [27], anchors fine-tuning to a prior, as transfer and continual learning do [18], [19], or averages a teacher, as semi-supervised learning does [20]. It has its own poisoning problem [17], [28], smaller when the attacker sees neither the online weights nor other users' samples [29], with robust-statistics defences [16]; a periodic reset to the pre-trained weights matches or betters continual adaptation over long streams [30]. These methods work without a physical check such as a code; the EATA- and CoTTA-style, anchored, teacher and median forms are compared below in receiver form. Byzantine-robust aggregation [31] needs many gradient vectors per step, certified defences against data poisoning [32] bound the loss of a convex learner, and a theory of continual learning under poisoning [33] treats regularisation-based learners over a sequence of tasks, with a defence that flags a task by comparing consecutive updates; here the learner is a network that sees one correlated block at a time. Nested learning [21] and Titans [22] learn memory at test time, the first at several update rates and the second in one long-term module written in proportion to a surprise measure; neither has a test that can refuse a write, and gated fast-to-slow consolidation echoes complementary learning systems [34], [35]. Adversarial examples against deep physical-layer models [36], [37], over-the-air poisoning of a wireless learner that re-trains on sensed data [38], and the methodology of attacks adapted to a defence [39]–[41] shape the evaluation. To the author's knowledge, no published deep receiver combines multi-rate memory, physics-gated consolidation and a retention bound.

The contributions are as follows.

1) **The principle and a receiver that realises it** (Section III): gated, clipped consolidation into two slow levels, a fallback that detects a rejected block with the slow memory, and per-sub-block CRC selection among the fast, slow and LMMSE decisions; the ablation identifies a slow level (the excess SER is $+0.316$ without one, $+0.0242$ with one and $+0.0199$ with two), the fallback and a decision-level gate as necessary, and attributes most of the clean-SER gain to the CRC selection.
2) **Retention bounds for the algorithm as implemented** (Section IV): a design bound, with no measured quantity, on the paired displacement of the slowest level under any attacker, and a structural bound for unequal gate verdicts computed from the paired trajectories. An error in the consolidation rule of an earlier version of this receiver (the first version below), which broke the convexity the bounds rely on, is corrected.
3) **Relative physics gates with a CRC design rule** (Section III-D): three label-free ratio tests and a design rule for the CRC threshold. Two tests compare the model residual and the CRC pass fraction of the fast level with those of the better of LMMSE and the anchor detector on the same block; the third compares the channel innovation with its running level. The rule sets the CRC threshold by a target excess symbol-error probability over that reference. The evaluated receiver uses the relative slack with a floor of $2{\cdot}10^{-3}$, which calibration chose over a slack derived from the pass-fraction statistics that follows the block length (Table VI); the derived rule is therefore an analysis, not what was evaluated.
4) **An evaluation with tuned baselines and attacks adapted to the defence** (Sections V–VI). The baselines, among them CRC-Rollback and anchored, decision-directed, EATA-style and CoTTA-style receivers, use the optimiser setting tuned for online DeepSIC and their own settings at default values, except the L2-SP weight, which is swept. The attacks are built against the defence (soft gates, deep unrolling, restarts, expectation over noise, joint pilot and data poisoning of the references, denial of adaptation, a slow small-budget attack optimised to be accepted, a grey-box surrogate and a query attacker that did not converge) and include a physical attacker with one, two or four antennas. The adapted attacks tested, all at full budget and rejected on nearly every attacked block, gave an excess not distinguishable from plain white-box poisoning (A1) with five seeds; the worst attack found without a channel change, a joint pilot and data attack (A4, Section VI-D) with 20 gradient steps, gave $+0.0478$, against $+0.0196$ for plain A1 on the same seeds. The scenarios include an NR LDPC link over 3GPP urban-micro drops with the attacker placed in the scene, 2000 clean blocks and, for plain

TABLE II
NOTATION.

U, N	users, base-station antennas
N_p, N_d, L, n	pilot and data symbols per user and block; symbols per CRC sub-block; sub-blocks per user and block
$\theta^{(k)}, K, C_k$	parameters of memory level k; number of levels; period of level k in blocks ($C_1 = 1$)
$\eta_b, \eta^{(k)}$	fast step size of block b; consolidation rate of level $k \geq 2$
a_b, λ, δ	anchor (slow projection); proximal weight; delta-rule decay
$\gamma_b, \tau_i, \omega, z_\mathrm{a}$	gate verdict of block b; threshold of gate Gi; soft-step width; abstain threshold
$F_\mathrm{nn}, F_\mathrm{ref}, p_\mathrm{ref}$	CRC pass fractions of the fast level and the reference; reference symbol-error probability
ρ, ξ, z, κ_k	relative CRC slack; excess symbol-error probability tolerated by G2; margin of the derived slack; consolidation clip of level k
$\boldsymbol{E}_b, \epsilon_b$	pilot perturbation; its budget
$W, n(W)$	window length in blocks; number of level-K consolidations in it
$D_k(b)$	paired displacement of level k after block b (attacked minus clean run; $k = 1$ is the fast level)

A1 only, a second network backbone and eight users with 32 antennas. Settings are fixed on calibration seeds by rules stated in advance, with two exceptions stated in Table IV (the abstain statistic and the channel profile of the larger arrays). Results are reported on held-out seeds with randomised attack windows, as the excess SER over a matched clean run with block-bootstrap intervals and seed-level tests. The failures that the evaluation exposes (reference degradation through the data, denial of adaptation during a channel change, per-user gating against a one-user attack, 16-QAM, the coded scene, the abstain state) are each reported with their mechanism.

Section II gives the system and threat model, Section III the receiver, Section IV the bounds, Section V the protocol, Section VI the results, Section VII their scope, and Section VIII concludes.

II. System Model and Threat Model

A. Uplink and Adaptive Receiver

U single-antenna users transmit QPSK symbols to a base station with N antennas. Time is divided into coherence blocks $b = 0, 1, \ldots$; within block b the channel $\boldsymbol{H}_b \in \mathbb{C}^{N \times U}$ is constant and the received vector at symbol slot n is

$$\boldsymbol{y}_n = f(\boldsymbol{H}_b \boldsymbol{s}_n + \boldsymbol{w}_n), \qquad \boldsymbol{w}_n \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}), \tag{1}$$

with unit-energy symbols, $\sigma^2 = \frac{1}{2}\, 10^{-\mathrm{SNR}/10}$ for a signal-to-noise ratio (SNR) in dB, and f the identity or the entrywise nonlinearity $f(x) = \tanh(0.5x)$ of [6], applied to the noisy received samples. Each block carries N_p pilot symbols per user, known to the receiver, and N_d data symbols framed in CRC-protected sub-blocks of L symbols per user. The main channel is the spatial exponential decay (SED) model of [5], $[\boldsymbol{H}_b]_{ij} = c_{j,b}\, e^{-|i-j|}$, with a per-user cosine fading gain $c_{j,b}$; Table V lists the other families. Table II collects the notation.

The receiver is DeepSIC [5]: Q soft-interference-cancellation iterations, each a per-user multilayer perceptron (MLP) that maps the received vector and the other users' soft decisions to a posterior over the user's symbol. Its parameters are adapted on the pilots of every block, $\theta_b = \Phi(\theta_{b-1}; \boldsymbol{Y}_b)$, by T steps of a first-order optimiser on the pilot cross-entropy, and the adapted networks then detect the data. Online DeepSIC starts every block from θ_{b-1}; Meta-DeepSIC [6] starts from an initialisation meta-learned by model-agnostic meta-learning (MAML) [42] on a buffer of the last 20 pilot blocks, refreshed every five blocks.

B. Threat Model

The adversary adds $\boldsymbol{E}_b$ to the received pilot matrix $\boldsymbol{Y}_b \in \mathbb{C}^{N_p \times N}$ (one row per pilot slot) of an attacked block, with $\|\boldsymbol{E}_b\|_F \leq \epsilon_b$ set by a signal-to-interference ratio (SIR),

$$\epsilon_b = \|\boldsymbol{R}_b\|_F \; 10^{-\mathrm{SIR}/20}, \tag{2}$$

where $\boldsymbol{R}_b = \boldsymbol{Y}_b$ (received energy, noise included) or $\boldsymbol{R}_b = \boldsymbol{S}_b \boldsymbol{H}_b^\top$ (users' signal energy only); both are reported. At SIR 0 dB the attacker's received pilot energy equals that of all U users together (plus the noise, with the first budget). As an order of magnitude, with a common path-loss exponent β, users at a common distance and equal transmit powers, the attacker must be $U^{1/\beta}$ times closer to the base station than the users: about 1.9 times for $U = 4$ and the line-of-sight urban-micro exponent $\beta = 2.1$ below the breakpoint distance [43]. The attacker is synchronised to the frame and knows the pilot sequence, which is standardised. The white-box attackers are given more than a transmitter can observe: they design $\boldsymbol{E}_b$ from the pilots as the base station receives them, noise included, and, with the soft-gate objective, from the users' data symbols and the received data of the same block (Table III); this bounds from above what a causal attacker can know. The attackers are:

- *A1, availability* [7]: projected gradient ascent (the PGD attack of [44]) on the post-adaptation cross-entropy of the clean received pilots, which share the block's channel and stand in for the data, through a differentiable surrogate of the adaptation map: plain gradient steps per stage (two of size $5 \cdot 10^{-3}$ unless stated) from the victim's current weights; Adam, the delta rule and the proximal pull are not unrolled.
- *A1 built against the defence*: deeper unrolling (10, 50 steps), more steps and random restarts, the soft-gate objective, and an expectation over noise draws (expectation over transformation, EOT [45]). The soft-gate objective is the data loss of the fast level, whose decisions the receiver emits if it accepts the block, times a differentiable surrogate of the probability that its gates accept, plus a reward for acceptance. The surrogate models the gates G1 (against LMMSE only) and G2 of Section III-D, not G3, with the defence's running statistics, the references of the unperturbed block, untightened thresholds and soft-step widths, a soft CRC pass rate and posterior-mean symbols; it replaces the hard gates on the forward and the backward pass, where the backward-pass differentiable approximation (BPDA) [39] keeps the exact forward pass.
- *A2, targeted*: A1 with the objective replaced by the post-adaptation loss of one user (user 1) minus that of the others; the perturbation covers the whole received pilot matrix.
- *A4, reference degradation*: a fraction α of the per-symbol power at the SIR on the pilots and $1-\alpha$ on the data, each budget set by (2) on its own part (total energy about

TABLE III
ATTACKER KNOWLEDGE.

White-box	victim's current weights, adaptation map, gate thresholds and running statistics; the block's received pilots (noise included) and noise power; for the soft-gate objective (also A3*, A4) the users' data symbols and the received data; differentiates through a gradient-descent surrogate of the adaptation map and, for the soft-gate objective, soft G1 and G2
Grey-box	the defence and its settings, not the weights: a Citadel-Rx surrogate trained on the pilots it observes
Transfer	an online DeepSIC surrogate trained on the pilots it observes
Query	no model: the NACK fraction of the victim's per-sub-block ACK/NACK feedback. One perturbation per attacked block, a kept direction plus a random probe, at the full budget; after the block the kept direction is moved along the probe in proportion to the NACK fraction against its running baseline, a one-sample search-gradient step after natural-evolution strategies [46], [47]. With one query per attacked block (40 in a window) the search takes few steps, so this row is not a converged query attack

$\alpha + (1-\alpha)N_d/N_p$ times that of A1, $N_d/N_p \approx 250$), optimised jointly for the soft-gate objective; the energy on the data also degrades the references the gates compare against.

- *Denial of adaptation*: a fixed channel-estimate spoof $\boldsymbol{E}_b = \boldsymbol{S}_b\boldsymbol{\Delta}^\top$, meant to make the gates reject every block, combined with a real abrupt channel change.
- *A3*, slow small-budget attack*: small perturbations over hundreds of blocks, of norm 5, 20 or 50% of that of the received pilots, each optimised for the soft-gate objective, so that its blocks are accepted and their updates are consolidated into the slow levels.
- *Physical attacker*: $\boldsymbol{E}_b = \boldsymbol{X}_b\boldsymbol{G}^\top$ with $M \in \{1,2,4\}$ attacker antennas and a fixed channel $\boldsymbol{G}$ with i.i.d. $\mathcal{N}(0, 1/N)$ entries drawn per seed; with an estimate $\hat{\boldsymbol{G}}$ of $\boldsymbol{G}$ whose error has relative variance 0.1 or 0.01, the attacker designs $\boldsymbol{E}_b = \boldsymbol{X}_b\hat{\boldsymbol{G}}^\top$ for $\hat{\boldsymbol{G}}$ and the receiver sees $\boldsymbol{X}_b\boldsymbol{G}^\top = \boldsymbol{E}_b(\boldsymbol{G}\hat{\boldsymbol{G}}^+)^\top$.
- *Attacker in a 3GPP scene*: the physical attacker placed in a TR 38.901 urban-micro (UMi) drop at 35, 75 or 150 m from the base station at the maximum power of a user, so that its SIR follows from the geometry, path loss and shadowing (Section VI-J).
- *Data jamming*: Gaussian noise at the same SIR on the data part, a control that separates persistent poisoning from plain interference.

Table III states what each attacker knows.

C. Metrics

The SER is the fraction of symbol slots in which at least one user is wrong [5]–[7]; the CRC is checked per user and sub-block, and the CRC gate uses the pass fraction over all users and sub-blocks (per user only with per-user gating). The primary metric is the *excess SER*: the SER of an attacked run minus that of the matched clean run, its *clean twin* (same configuration, seed and channel draws, attack disabled, same machine), averaged over the attack window. The two runs of a pair are identical up to the first attacked block, so the excess isolates the attack from the difficulty of the blocks it hits. The gate is characterised by the rejection rate of poisoned blocks and the false rejection rate of clean blocks. The tables give rates; the text gives counts for some of them and a Wilson interval [48] at the headline cell, which is white-box A1 at SIR 0 dB in the main configuration of Table V. Unless a caption states otherwise, a rejection rate over s seeds is a fraction of the attacked blocks of the s runs ($40s$ for a 40-block window) and a false-rejection rate a fraction of the blocks of their clean twins from block 5 on ($95s$ for 100-block runs). The paired displacement $D_k(b) = \left\|\theta^{(k)}_{\mathrm{adv},b} - \theta^{(k)}_{\mathrm{clean},b}\right\|$ measures what the attack changed in level k, whether it pushed the level or held it back.

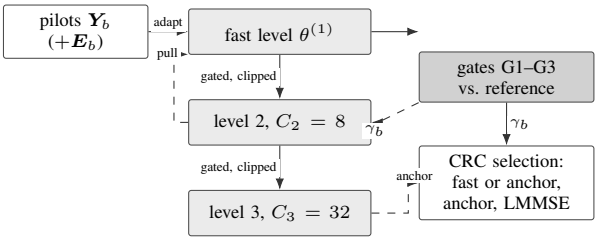


Fig. 1. Citadel-Rx. The fast level fits every block; the gates compare its decisions with the better of LMMSE on the block's pilots and the anchor detector; levels 2 and 3 receive gated, clipped averages of the faster states; a rejected block is reverted and detected with the anchor; every CRC sub-block keeps the first passing detection. The anchor is the mean of levels 2 and 3; its two arrows are drawn from one level each.

III. CITADEL-RX

The receiver splits adaptation by time-scale: a fast level fits every block; slow levels change only through what the gates accept (Fig. 1, Algorithm 1).

A. Levels and the Fast Learner

The adaptable parameters $\theta \in \mathbb{R}^d$ (all DeepSIC weights, $d = 67\,632$ for 4×4 QPSK) are kept at $K = 3$ levels with periods $C = (1, 8, 32)$ blocks. The fast level is adapted on every block by $T = 300$ steps of

$$\theta \leftarrow \theta - \eta_b \delta\, \theta\, \boldsymbol{\phi}\boldsymbol{\phi}^\top \quad \text{(delta-rule decay, per weight matrix)}, \tag{3}$$

$$\theta \leftarrow \theta - \eta_b \lambda(\theta - a_b) \quad \text{(proximal pull to the anchor)}, \tag{4}$$

$$\theta \leftarrow \mathrm{Adam}\big(\theta, \nabla_\theta \ell_b(\theta)\big), \tag{5}$$

with ℓ_b the pilot cross-entropy of the stage being adapted, its gradient evaluated before (3) and (4) are applied (stages are adapted sequentially [5]), $\boldsymbol{\phi}$ the normalised mean input of the weight matrix (the stage input for the first layer, the hidden activation for the second; biases are not decayed), and $a_b = \frac{1}{2}\theta^{(2)}_{b-1} + \frac{1}{2}\theta^{(3)}_{b-1}$ the anchor (weights $w_k = \frac{1}{2}$ in Algorithm 1). The step size follows the ratio of the receiver's surprise $S_b = \left\|\nabla_\theta \ell^{\mathrm{det}}_b(\theta_{b-1})\right\|^2$ to its moving average $\bar{S}_b$, where ℓ^{det}_b is the pilot cross-entropy of the posteriors of the full detector: $\eta_b = \eta_{\max}\, 2r_b/(1+r_b)$ with $r_b = \mathrm{clip}(S_b/\bar{S}_b, \frac{1}{8}, 8)$, so the rate is bounded by $2\eta_{\max}$.

B. Gated, Clipped Consolidation

At the end of every period of level $k{+}1$ the level receives the gated mean of the level-k states buffered since its last

update, each weighted by the verdict $\gamma_j \in [0,1]$ of its block (or the mean verdict of its period):

$$\theta^{(k+1)} \leftarrow \theta^{(k+1)} + \Pi_{\kappa_{k+1}}\Big(s\big(m-\theta^{(k+1)}\big)\Big),$$
$$m = \frac{\sum_j \gamma_j \theta_j^{(k)}}{\sum_j \gamma_j}, \qquad s = \eta^{(k+1)} \min\big(1, \textstyle\sum_j \gamma_j\big), \tag{6}$$
$$\Pi_\kappa(\boldsymbol{v}) = \boldsymbol{v}\min\{1, \kappa/\|\boldsymbol{v}\|\},$$

with $\eta^{(2)} = 1$, $\eta^{(3)} = \frac{1}{2}$, and no update when $\sum_j \gamma_j = 0$. The clip $\kappa = (8.5, 4.5)$ lies at or just above the largest clean consolidation increments of levels 2 and 3 on the calibration runs (8.2 and 4.5), so it rarely binds on clean runs (in 5/150 consolidations of the clean twins at the headline cell) and limits what any attacked period can move. The first version of the receiver divided the gated sum by $\max(1, \sum_j \gamma_j)$; when the verdicts of a period sum to less than one, that is, when nearly all its blocks are rejected, which is the case under attack, this pulls the level towards the origin instead of leaving it nearly unchanged, and the update is then not the convex combination that the bounds of Section IV require. Every result of this paper uses (6).

C. Rejection, Fallback and CRC Selection

If $\gamma_b < \frac{1}{2}$ the block is rejected: the fast update is reverted, the reverted state enters the buffer with weight γ_b, and the data of the block are detected with the anchor, which the pilots of this block have not changed; attacked blocks that were accepted earlier have entered it through the consolidation (Section VI-E). Each CRC sub-block of each user then keeps the first of three detections that passes its CRC: the gate's choice (fast level or anchor), the anchor, and LMMSE on the least-squares channel estimate. All three are already computed for the gates; when none passes, the gate's choice is kept.

Four optional mechanisms answer specific attacks and are evaluated with them (Sections VI-D–VI-E). *Abstain*: the block is rejected and not buffered when the reference itself has degraded. The first version tested the reference's pass fraction and residual against their running levels on accepted blocks ($z_\mathrm{a} = 4$, binomial floor on the variance), and on clean runs it tripped and then rejected every later block (Section VI-D). The final version tests $t_b = \log(r_b^\mathrm{v}/\hat{\sigma}_b^2)$, where r_b^v is the mean squared residual $\left\|\boldsymbol{y}_n - \hat{\boldsymbol{H}}_b \boldsymbol{s}_n\right\|^2/N$ over the symbol slots whose decisions the CRC verifies for every user (the residual of the LMMSE decisions if fewer than 5% are verified) and $\hat{\sigma}_b^2$ is the noise power estimated on the pilots, both per antenna. Without attack both follow the noise power (the data residual also carries the channel-estimation error, and the verified slots leave out the largest noise samples, two offsets that the running level tracks), whatever the channel, because verified decisions are correct; poisoned pilots inflate $\hat{\sigma}_b^2$ and energy on the data inflates r_b^v. The receiver abstains when $|t_b - \bar{t}_b| > z_\mathrm{a}\max(s_b, 0.1)$, with the level $\bar{t}_b$ and the spread s_b tracked on every non-abstained block, and re-bases the level after 50 consecutive abstentions, so that a lasting change does not make it abstain indefinitely; in the runs with a change at block 50 of 100 the re-base can fire only at the last block, so it is not evaluated. An intermediate version on the residual of the hard LMMSE decisions was dropped after its first test runs: that residual also follows the decision errors of the reference, which a legitimate redraw to a worse-conditioned channel raises. *Per-user gating*: G2 is evaluated per user with that user's pass fractions, and only the users whose test fails revert their networks and fall back to the anchor (G1 and G3 still judge the block). *Streak recovery*: after five consecutive rejections a re-fit from the anchor on CRC-verified data, kept if it verifies better than the reference. *Drift guard*: a pull of level K towards the pre-trained weights by 5% per consolidation.

Algorithm 1 Citadel-Rx, one block b

Require: pilots $\boldsymbol{Y}_b$ (possibly poisoned), data $\boldsymbol{Y}_b^d$; levels $\theta^{(1..K)}$; buffers; running statistics
$a_b \leftarrow \sum_{k\geq 2} w_k \theta^{(k)}$; $\eta_b \leftarrow \eta_\mathrm{max} 2r_b/(1+r_b)$
reference: $\hat{\boldsymbol{H}}_b$ (LS, AR(1) tracker); $\hat{\boldsymbol{s}}_\mathrm{mm}$ (LMMSE); $\hat{\boldsymbol{s}}_a$ (anchor)
$\theta \leftarrow \theta^{(1)}$; T steps of (3)–(5) on $\boldsymbol{Y}_b$; $\hat{\boldsymbol{s}}_\mathrm{nn} \leftarrow$ detect$(\theta, \boldsymbol{Y}_b^d)$
$\gamma_b \leftarrow \min(\gamma_1, \gamma_2, \gamma_3)$: G1 residual ratio, G2 CRC ratio $F_\mathrm{nn}/F_\mathrm{ref}$, G3 innovation
if $\gamma_b < \frac{1}{2}$ **then**
  $\theta \leftarrow \theta^{(1)}$ (revert); $\hat{\boldsymbol{s}} \leftarrow \hat{\boldsymbol{s}}_a$ (fallback)
else
  $\hat{\boldsymbol{s}} \leftarrow \hat{\boldsymbol{s}}_\mathrm{nn}$; update the running statistics
end if
per user and sub-block: $\hat{\boldsymbol{s}} \leftarrow$ first CRC-passing of $(\hat{\boldsymbol{s}}, \hat{\boldsymbol{s}}_a, \hat{\boldsymbol{s}}_\mathrm{mm})$
$\theta^{(1)} \leftarrow \theta$; buffer $(\gamma_b, \theta^{(1)})$; for every k with $(b+1) \bmod C_{k+1} = 0$: (6)
return $\hat{\boldsymbol{s}}$

D. Relative Physics Gates

The statistics of G1 and G2 are ratios against a reference computed on the same block without labels: the better of the LMMSE detector on the least-squares (LS) pilot estimate $\hat{\boldsymbol{H}}_b$ and the anchor detector. The statistic of G3 is a ratio against its own running level. The ratios are normalised by their moving averages over accepted blocks and passed through a soft step of width $\omega = 0.1\tau$ at the threshold τ, which is tightened by up to 30% when $r_b > 1$.

G1, model residual. With $r(\hat{\boldsymbol{s}}) = \frac{1}{N_d}\sum_n \left\|\boldsymbol{y}_n - \hat{\boldsymbol{H}}_b \hat{\boldsymbol{s}}_n\right\|^2$, the statistic is $r(\hat{\boldsymbol{s}}_\mathrm{nn})/r(\hat{\boldsymbol{s}}_\mathrm{ref})$ against $\tau_1 = 1.25$.

G2, CRC pass fraction. The statistic is the pass fraction F_nn of the fast level over that of the reference, F_ref, against the threshold τ_2 of the design rule of Section IV. The gate is tied to the availability attacker's own objective: an attack that raises the SER lowers the pass fraction. With LDPC code blocks (Section VI-J) the CRC after decoding tolerates raw symbol errors, so the pass fraction of a code block does not follow the design rule. There F is replaced by the virtual pass fraction $(1-\hat{p})^L$, and the rule applies unchanged. Here $\hat{p}$ (the p of the design rule) is the symbol error rate of a detector over all users (per user with per-user gating), measured on the code blocks that decode under a reference, whose re-encoded codewords are taken as the transmitted symbols. When no code block decodes under any reference, the code-block pass fractions are used.

G3, channel prior. The statistic is the innovation energy of the LS estimate with respect to its first-order autoregressive, AR(1), prediction (the previous block's estimate, scaled by the channel's correlation on Jakes fading), divided by its running level on accepted blocks, which is floored by its expected value. It is tested against $\tau_3 = \max\{4, \chi^2_{0.999}(NU)/(NU)\}$, which is 4 for every array here.

TABLE IV
SETTINGS AND THEIR PROVENANCE. CALIBRATION: SEEDS 0–2 BY THE RULES OF SECTION V; FIRST VERSION: SET DURING THE DEVELOPMENT OF THE FIRST VERSION ON SEEDS 0–4. NO SETTING WAS CHOSEN ON THE TEST SEEDS 10–19, WITH TWO EXCEPTIONS: THE STATISTIC OF THE FINAL ABSTAIN STATE, ADOPTED AFTER FIRST TEST RUNS OF ITS SECOND VERSION (STATED IN THE TABLE), AND THE SED PROFILE OF THE LARGER ARRAYS, STRETCHED AFTER TEST RUNS WITH THE UNSTRETCHED ONE (SECTION VI-H).

Setting	Value; provenance
Q, MLP width, T, $\eta_{\max}$, Adam	3, $64\cdot$order for QPSK ($=256$), 300, $5\cdot10^{-3}$; reference code released with [49], architecture of [5]; width kept at 256 for 16-QAM (not $64\cdot16$, Section VI-H)
K, C	3, $(1,8,32)$; first version; ablation in Section VI-G
$\eta^{(2)}, \eta^{(3)}$, λ, δ	1, $\frac{1}{2}$, 0.5, 0.1; first version; sensitivity of $\eta^{(3)}$, λ, δ in Section VI-K
τ_1, τ_3, ω, tightening	1.25, 4, 0.1τ, 30%; first version; sensitivity
Surprise clip, rejection threshold, anchor weights, G3 quantile	$[\frac{1}{8}, 8]$, $\frac{1}{2}$, $(\frac{1}{2}, \frac{1}{2})$, 0.999; first version
CRC slack	relative $\rho=1$ with floor $2\cdot10^{-3}$ (first version), or derived $z=5$ (set before calibration); mode chosen by calibration (relative slack)
κ	$(8.5, 4.5)$; set before calibration from the clean consolidation increments of earlier records of the calibration configurations; common to all candidates
Consolidation rule, CRC selection	(6); correctness fix; CRC selection common to all calibration candidates (ablation in Section VI-G)
Baseline epochs, learning rate, weight decay	tuned on calibration seeds (Section V)
Baseline method settings	L2-SP, EWC 10^{-2}; EMA 0.9; CoTTA restore 0.05; EATA $0.4\ln4$; decision-directed weight 1; drift threshold 2; four median-of-means groups; not tuned (L2-SP also swept, Section VI-F)
Streak k, drift guard, abstain fallback, spread floor	5, 5%, 5% verified, 0.1; one value each, not tuned
Abstain (final version)	statistic and re-base after 50 abstentions: adopted after the second version's first test runs (seeds 10–12, A4); $z_{\mathrm{a}}=3$ by calibration; also tested on seeds 15–19, untouched by any choice
Second backbone	512 hidden units; epochs and learning rate tuned on calibration seeds (30 epochs, learning rate 0.001)
Attacker	PGD 20 steps of 0.25ϵ, 2 unrolled steps; our implementation of the pilot poisoning of [7]; stronger variants in Section VI-C

Table IV gives the provenance of the main settings.

IV. RETENTION BOUNDS

The bounds are stated for the algorithm as implemented, for a pair of runs on the same channel, noise and seed, with and without an attack that starts at block b_0 (the runs are identical before b_0). Let $n(W)$ be the number of level-K consolidations in $[b_0, b_0+W)$, so $n(W)\le\lceil W/C_K\rceil$; quantities of the clean run are primed.

Lemma 1 (convex consolidation)**.** *Without the clip,* (6) *writes the new level-$(k+1)$ state as the convex combination $(1-s)\theta^{(k+1)}+s\,m$ of its old state and the buffered states (for $\sum_j\gamma_j>0$), with $s\le\eta^{(k+1)}\le1$ and m in their convex hull. With the clip the step is $\tilde s = s\min\{1,\kappa_{k+1}/\left\|s(m-\theta^{(k+1)})\right\|\}\le s$, and the statement holds with $\tilde s$.*

Theorem 1 (retention)**.** *(i)* Design bound. *With the clip, for any perturbations and any gate verdicts,*

$$D_K(b_0+W-1)\le 2\kappa_K\, n(W). \tag{7}$$

(ii) Structural bound. *Let the two runs buffer the same blocks (always the case without the abstain state) and let $\sum_j\gamma_j>0$ in both at every level-K consolidation of the window. Let $\Delta(W)$ be the largest paired displacement of the buffered level-$(K-1)$ states in the window and, at the i-th consolidation, $\bar w_i, \bar w_i'$ the normalised weights, $\tilde s_i, \tilde s_i'$ the steps taken, m_i' the clean gated mean and r_i' the radius of the clean buffer around it, and $\theta_i'^{(K)}$ the clean level-K state before that consolidation. Then*

$$D_K(b_0+W-1)\le\big(\Delta(W)+\Omega(W)\big)\big(1-(1-\eta^{(K)})^{n(W)}\big) + \sum_{i\le n(W)}|\tilde s_i-\tilde s_i'|\,\left\|\theta_i'^{(K)}-m_i'\right\|, \tag{8}$$

with $\Omega(W)=\max_i\|\bar w_i-\bar w_i'\|_1\,r_i'$. With equal verdicts $\Omega=0$; if moreover the clip binds in neither run, or with the same factor in both, at every consolidation of every level, then $\tilde s_i=\tilde s_i'$, the sum vanishes, and $\Delta(W)$ is at most the largest paired fast-level displacement.

Proof. (i) Level K changes only at consolidations, and in each run by at most κ_K; the drift guard, when used, multiplies the paired difference by 0.95. (ii) By Lemma 1, at consolidation i, $\theta_{\mathrm{new}}^{(K)}-\theta_{\mathrm{new}}'^{(K)}=(1-\tilde s_i)(\theta^{(K)}-\theta'^{(K)})+\tilde s_i(m_i-m_i')+(\tilde s_i-\tilde s_i')(m_i'-\theta'^{(K)})$. Since $\sum_j\bar w_{ij}=\sum_j\bar w_{ij}'=1$, $m_i-m_i'=\sum_j\bar w_{ij}(\theta_j-\theta_j')+\sum_j(\bar w_{ij}-\bar w_{ij}')(\theta_j'-m_i')$, whose norm is at most $\Delta+\Omega$. Iterating $D\leftarrow(1-\tilde s_i)D+\tilde s_i(\Delta+\Omega)+e_i$ from $D=0$ gives $D\le(\Delta+\Omega)(1-\prod_i(1-\tilde s_i))+\sum_i e_i$, and $\tilde s_i\le\eta^{(K)}$; as in (i), the drift guard only shrinks the paired difference. □

Bound (7) needs no measured quantity and holds for every attacker of this paper, including those that pass the gates; its price is that a legitimate change of the channel moves the slowest level by at most κ_K per period. The same argument gives $D_2\le2\kappa_2\lceil W/C_2\rceil$ for $K=3$, so the anchor, which the fallback and the proximal pull use, is displaced by at most $\kappa_2\lceil W/C_2\rceil+\kappa_K n(W)$ after the last block of the window. Bound (8) is evaluated on the paired trajectories in Section VI-E; it is tighter when the verdicts agree, and it is the form in which the gate enters: rejected blocks are reverted and enter with weight γ_b.

Remark 1 (contraction). If the block map satisfied $\|\Phi(\theta;\boldsymbol{Y}+\boldsymbol{E},a)-\Phi(\theta;\boldsymbol{Y},a)\|\le G_\Phi\,\|\boldsymbol{E}\|$ and were jointly L_Φ-Lipschitz in (θ,a) with $L_\Phi<1$, and if every block of the window were accepted in both runs with equal verdicts and equal clip factors, so that by Lemma 1 the paired displacement of the anchor is at most the largest paired fast displacement so far, the largest fast displacement Δ_f after n_W attacked blocks of budget ϵ would satisfy $\Delta_f\le G_\Phi\epsilon(1-L_\Phi^{n_W})/(1-L_\Phi)$; on a block accepted in one run only, the displacement can grow by the norm of that run's update, which does not scale with ϵ. The sufficient condition is $L_\theta+L_a<1$, with L_θ and L_a the Lipschitz

constants of Φ in θ and in a, whose sum bounds L_Φ (taken for $\max\{\|\theta-\theta'\|, \|a-a'\|\}$) from above. Finite-difference probes of the fast learner, which the correction of (6) does not change, give $L_\theta + L_a$ between 1.08 and 1.39 for proximal weights λ from 0.5 to 4, and a stronger search, a finite-difference hill climb on the input direction for G_Φ and the largest of 20 random parameter and anchor directions on each of 10 blocks for L_θ and L_a, gives 1.03 to 2.01 on the cosine, Jakes and COST2100 channels at the frozen settings (with G_Φ of at least 10.72, 4.50 and 2.06 on the three); probes can only underestimate a Lipschitz constant, so the sufficient condition is not met, and on the cosine channel the probe of L_θ alone (1.23) exceeds one, which excludes $L_\Phi < 1$; the contraction argument cannot be used for the default receiver. The regret bound of the first version, which relied on it, is withdrawn; a last-layer-adaptable variant, for which the loss is convex, is evaluated in Section VI-H without a regret claim.

Design rule for the CRC gate. With i.i.d. symbol errors of probability p in a sub-block of L symbols and an r-bit CRC whose syndrome is uniform on error, $P(\text{pass} \mid p) = (1-p)^L + 2^{-r}(1-(1-p)^L)$. Neglecting the 2^{-r} term, inverting at F_{ref} gives $p_{\text{ref}} = 1 - F_{\text{ref}}^{1/L}$, and the rule "reject if $F_{\text{nn}}/F_{\text{ref}} < \tau_2$" with $\tau_2 = \big((1-p_{\text{ref}}-\xi)/(1-p_{\text{ref}})\big)^L$ tolerates an excess error probability ξ over the reference. The gate applies the rule to the ratio normalised by its moving average over accepted blocks (Section III-D), so that, to first order, ξ is counted from the fast level's usual error probability, which is that of the reference only when the moving average is one. The relative slack sets $\xi = \max\{2\cdot 10^{-3}, \rho\, p_{\text{ref}}\}$; the derived slack removes the floor and takes $\tau_2 = \min\{1 - z\sigma, \tau_2(\rho p_{\text{ref}})\}$, with σ the standard deviation of the normalised pass ratio on accepted blocks, floored by the binomial relative standard deviation $\sqrt{F_{\text{ref}}(1-F_{\text{ref}})/n_{\text{c}}}/F_{\text{ref}}$ of the reference pass fraction over the n_{c} CRC checks of the statistic ($n_{\text{c}} = nU$ for the block gate, $n_{\text{c}} = n$ per user), so the threshold follows the block length. For n independent sub-blocks, a pass fraction $\hat{F}$ with expectation F satisfies $P(|\hat{F} - F| \geq t) \leq 2e^{-2nt^2}$ [50]. The formula is standard; its use here is to replace a tuned threshold by a target excess error. In the evaluated receiver that target keeps the first-version values ($\rho = 1$, floor $2 \cdot 10^{-3}$), which were not derived. Here p is the symbol-error probability of a user, pooled over the users by the gate; the SER of Section II counts a slot in which any user is wrong and is therefore at most U times this p. The independence of the symbol errors within a sub-block is an assumption that was not checked against the records.

Model mismatch and G1. Writing $\boldsymbol{y}_n = \hat{\boldsymbol{H}}\boldsymbol{s}_n + \boldsymbol{\nu}_n + \boldsymbol{m}_n$ with $\boldsymbol{m}_n$ the mismatch, the residual of a decision with error $\boldsymbol{e}_n$ is $\text{mean}\left\|\hat{\boldsymbol{H}}\boldsymbol{e}_n\right\|^2 + \text{mean}\,\|\boldsymbol{\nu}_n + \boldsymbol{m}_n\|^2$ plus a cross term that vanishes if decision errors are uncorrelated with noise and mismatch. This is an assumption, not a fact: errors occur when the noise is large. On clean blocks of the $\tanh$ channel at 10 dB (first-version runs, seeds 0–2) the normalised G1 statistic has mean 1.000, 99th percentile 1.050 and maximum 1.080 over 285 blocks; its largest value at any SNR from 4 to 16 dB is 1.194, below $\tau_1 = 1.25$ but above the fully tightened threshold $1 + 0.7(\tau_1 - 1)$, so G1 can reject a clean block of this channel when the surprise is high.

V. Experimental Protocol

Benchmark channel. The SED channel of the DeepSIC literature (Table V) is real-valued with deterministic fades, identical for every seed: the seeds change the noise, the symbols and the attack window, and pre-training uses its block-0 matrix at several SNRs. The test seeds therefore do not test a new channel.

Receivers. The receivers compared are LMMSE on the least-squares pilot estimate and a Huber-robust version [51]; online DeepSIC [5]; Meta-DeepSIC [6]; *CRC-Rollback*, online DeepSIC with the CRC gate G2 at a fixed absolute slack of 0.01 ($\tau_2 = 0.99^L$, tightened under surprise as in Section III-D), judged against LMMSE only, without CRC selection, and with rollback to the last accepted weights, and the same with all three gates (the $K = 1$ ablation, Section VI-G, has Citadel-Rx's gates and selection); online DeepSIC with the CRC selection of Citadel-Rx, which separates the selection gain from the memory; L2-SP [18], elastic weight consolidation (EWC) with the Fisher information of the pre-training data [19] and an exponential-moving-average (EMA) teacher [20]; CRC-verified decision-directed training, which adds the CRC-passing data symbols to the pilots [10], [11]; an EATA-style receiver (entropy filter and Fisher anchor) [12] and a CoTTA-style receiver (teacher and stochastic restore) [13]; a drift-detection receiver after [25], with a simpler statistic, that adapts a user's modules only when that user's pilot loss, standardised by its running mean and spread, exceeds 2; a median-of-means gradient receiver (coordinate-wise median of the gradients of four disjoint pilot groups), which moves the median of [16] from the batch-normalisation statistics, which DeepSIC does not have, to the gradient; and a single-memory receiver with the fast learner of Citadel-Rx and no slow levels [22]. All DeepSIC receivers share the architecture and the pre-training. A second backbone replaces DeepSIC in Citadel-Rx, online training and CRC-Rollback: a fully connected network with one hidden layer of 512 units that classifies the joint symbol of all users ($4^U = 256$ classes) from the received vector, without interference cancellation [1], [3]; detection uses its per-user marginals.

Calibration and tuning, fixed in advance. Online DeepSIC and Meta-DeepSIC are tuned over 300, 100 and 30 adaptation epochs, learning rates 10^{-3} and $5 \cdot 10^{-3}$ and weight decay 0 and 10^{-4} on clean runs of the calibration seeds; the setting with the lowest mean clean SER is selected (online DeepSIC: 30 epochs, learning rate 0.001, weight decay 0; Meta-DeepSIC: 300, 0.001, 0), and every receiver built on online DeepSIC uses its setting (the drift-detection and median-of-means receivers, first run at the reference 300 epochs and learning rate $5 \cdot 10^{-3}$, were re-run with it for the tables); the baselines' own method settings are not tuned (Table IV). Citadel-Rx has three candidates: the relative CRC slack, the derived CRC slack, and the relative slack with a fast level that is restarted from the anchor every block and whose update per block is projected onto a ball of radius 13. All three use

TABLE V
SIMULATION SETUP.

System	$U \times N = 4 \times 4$, QPSK, SNR 10 dB; 4×16 and 8×32 with the SED profile stretched over the array; 16-QAM at 20 dB (MLP width kept at 256)
Block	$N_p = 100$ pilot and $N_d = 24\,992$ data symbols per user; $N_d = 992$ and 4 992 (nominally 1000 and 5000), $N_p \in \{32, 100\}$ in Section VI-H
CRC	CRC-8 of [52], $D^8 + D^7 + D^4 + D^3 + D + 1$; $L = 32$; $n = 781$ sub-blocks per user and block
Channel	SED with per-user cosine fading, gain $0.8 + 0.2\cos(2\pi b/T_j)$ with $T_j = 51, 39, 33, 21$ blocks (users 5–8: 47, 29, 43, 23) [5]; COST2100 traces [53] from the code released with [49], in calibration
Run	100 blocks (400 and 1000 for A3*, 2000 clean); unless stated, a 40-block attack window starting at a block drawn uniformly from 10–50 per seed; fixed windows: 30–69 for calibration and retention, 40–89 for denial, 30–329 and 30–929 for A3*
Seeds	calibration 0–2; test 10–19 (headline cell, final abstain state), up to 10–14 elsewhere (every table states its seeds), 10 (sensitivity)
Coded scene	TR 38.901 UMi at 3.5 GHz, 1×4 array at 10 m, users at 20–150 m with ideal slow power control, 3 km/h; one block per 1 ms slot (15 kHz, 150 PRBs); NR LDPC code blocks (base graph 2, $k = 520$, $n = 1000$), each with a CRC-24A [54], scrambled [55]
Attacker	A1, after [7]: PGD [44], 20 steps of 0.25ϵ, through 2 unrolled steps, unless stated

the CRC selection, the clip κ and (6). The rule, fixed before the runs, selects the candidate with the lowest mean of the clean SER, the A1 window SER at SIR 0 dB and the A1 window SER on COST2100, among candidates with at most 2% clean false rejections (Table VI). The chosen settings are written to a file by the rule and read by every test run. The second campaign added two choices, again on seeds 0–2 by rules fixed before any test run of the features: the abstain threshold z_a is the smallest of 3, 4, 5, 6 and 8 with at most 2% abstentions on the clean blocks of calibration runs that never abstain (285 blocks; chosen $z_a = 3$, largest deviation 1.8), and online training on the second backbone is tuned over 30 and 100 epochs and learning rates 10^{-3} and $5 \cdot 10^{-3}$ like online DeepSIC (30 epochs, learning rate 0.001); Citadel-Rx keeps all its frozen settings on the second backbone and in the coded scene.

Statistics. Within a run, consecutive blocks are correlated (a decorrelation length of 6 blocks for the excess SER at the headline cell), so intervals are 95% moving-block bootstrap intervals [56] (percentile intervals from 4000 resamples) that resample seeds and then, within each seed, stretches of consecutive blocks of twice the decorrelation length (at least five blocks), wrapped around the end of the series. Tests are at the seed level: an exact two-sided Wilcoxon signed-rank test on the per-seed means [57], whose smallest attainable p with ten seeds is 0.002, Holm-corrected over the four baselines of a column [58]. Rejection and false-rejection rates pool the blocks of all seeds of a cell; their Wilson intervals at the headline cell [48] treat these blocks as independent and are too narrow if the verdicts of consecutive blocks are correlated.

Reproducibility. The first campaign ran on an Apple M4 laptop (ten cores, eight runs at a time) and a 4-core cloud machine, the second on 4-core cloud machines, every run single-threaded, with disjoint sets of runs. Runs used PyTorch 2.14.1, NumPy 2.5.3 and SciPy 1.18.1 with one CPU thread per run; each attacked run and its clean twin ran on one machine (or, for re-runs, on a machine that reproduces the twin block for block before the window). Times per block come from the M4 runs

TABLE VI
CALIBRATION OF CITADEL-RX ON SEEDS 0–2 (SER). SCORE: MEAN OF THE CLEAN, A1 AND COST2100 A1 COLUMNS.

Candidate	Clean	A1 window	COST2100 A1	False rej.	Score	
Relative slack	0.0151	0.0245	$2.8 \cdot 10^{-4}$	0.000	0.0133	chosen
Derived slack	0.0150	0.0272	$2.3 \cdot 10^{-4}$	0.000	0.0141	
Relative slack, anchor start	0.0168	0.0318	$2.2 \cdot 10^{-4}$	0.000	0.0163	

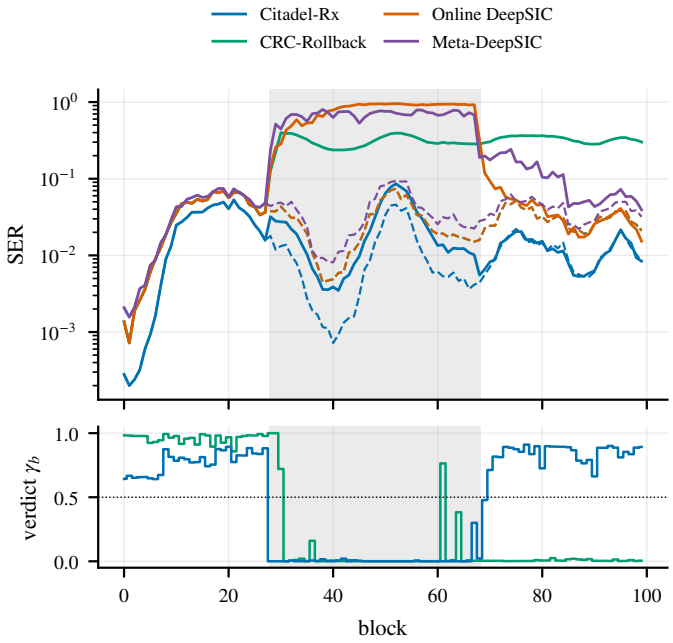


Fig. 2. Test seed 10, white-box A1 at SIR 0 dB in the shaded window. Top: SER per block of the attacked runs (solid) and their clean twins (dashed). Bottom: gate verdict γ_b of CRC-Rollback and Citadel-Rx; a block is rejected below $\frac{1}{2}$.

and compare receivers only roughly; machine-independent costs are counted in multiply-accumulates. Every record stores its configuration, seed, code version and timings. Pre-training is cached by configuration, seed, code and random-number state, and a cached run reproduces the uncached run exactly (the repository's tests compare the two). Three commands re-run the two campaigns, and `make paper` recomputes every number, table and figure of this paper from the records and rebuilds it (README, section Reproducibility).

VI. RESULTS

A. Headline Comparison

Table VII contains the main result. Under white-box A1 at SIR 0 dB the attack raises the SER of Citadel-Rx in the attack window by $+0.0188$ over its clean twin (95% interval [0.0142, 0.0246]; seed means from $+0.0130$ to $+0.0279$), and that of tuned online DeepSIC by $+0.768$. The excess of Citadel-Rx is small but not zero: it is positive on 10 of 10 seeds (exact Wilcoxon $p = 0.002$). The gates reject 397/400 poisoned blocks (95% Wilson interval [0.978, 0.997]) and 1/950 clean blocks of the clean twins ([0.000, 0.006]). Citadel-Rx also has the lowest excess at SIR -10 and $+10$ dB and under

TABLE VII
EXCESS SER (ATTACKED MINUS MATCHED CLEAN RUN, ATTACK WINDOW) AT THE HEADLINE CELLS, TEST SEEDS; 95% MOVING-BLOCK INTERVAL AT WHITE-BOX SIR 0 DB. SEEDS 10–19 FOR WB 0 DB, CLEAN, REJ. AND FALSE REJ., SEEDS 10–14 FOR THE OTHER COLUMNS. CLEAN: SER OF THE CLEAN TWINS OVER BLOCKS 5–99. REJECTION OF THE POISONED BLOCKS AND FALSE REJECTION ON THE CLEAN TWINS, BY THE GATES OF CITADEL-RX AND OF CRC-ROLLBACK, AT WHITE-BOX SIR 0 DB.

Receiver	Clean	WB −10 dB	WB 0 dB	WB +10 dB	Transfer 0 dB	Jamming 0 dB	Rej.	False rej.
Citadel-Rx	0.0159	+0.0216	+0.0188 [0.0142, 0.0246]	+0.0118	+0.0195	+0.835	0.993	0.001
Online DeepSIC	0.0330	+0.876	+0.768 [0.715, 0.815]	+0.294	+0.399	+0.818	–	–
Meta-DeepSIC	0.0393	+0.900	+0.651 [0.630, 0.670]	+0.204	+0.205	+0.813	–	–
CRC-Rollback	0.0330	+0.513	+0.271 [0.234, 0.307]	+0.0523	+0.253	+0.820	0.897	0.000
Online + CRC selection	0.0209	+0.888	+0.765 [0.711, 0.814]	+0.182	+0.394	+0.830	–	–

TABLE VIII
PAIRED SEED-LEVEL COMPARISON, CITADEL-RX MINUS BASELINE: ATTACK-WINDOW SER AT WHITE-BOX SIR 0 DB AND CLEAN SER (RELATIVE CHANGE), EXACT WILCOXON p, HOLM-CORRECTED OVER THE FOUR BASELINES.

Baseline	Seeds	Δ window SER	p	Δ clean SER	p
Online DeepSIC	10	−0.765	0.008	−0.0171 (−52%)	0.008
Meta-DeepSIC	10	−0.656	0.008	−0.0234 (−60%)	0.008
CRC-Rollback	10	−0.269	0.008	−0.0171 (−52%)	0.008
Online + CRC selection	10	−0.750	0.008	−0.0051 (−24%)	0.008

transfer (5 seeds each); the closest baseline, CRC-Rollback at +10 dB, has +0.0523 against +0.0118. Over the blocks after the window (30, or fewer when the window ends late in the run) the excess of Citadel-Rx is $+3.8 \cdot 10^{-4}$; online DeepSIC keeps +0.0189, CRC-Rollback +0.346, and Meta-DeepSIC, whose meta-training buffer keeps poisoned pilots for 20 blocks after the attack, +0.0882. Fig. 2 shows one seed.

Paired seed by seed (Table VIII), Citadel-Rx has the lower window SER against each of the four baselines on every seed (Holm-corrected $p = 0.008$, the smallest attainable with ten seeds and four tests), and also the lower clean SER, 52% below that of tuned online DeepSIC. Most of the clean gain, 70%, comes from the per-sub-block CRC selection rather than from the memory: online DeepSIC with the same selection has a clean SER of 0.0209, against 0.0330 without it and 0.0159 for Citadel-Rx, and the selection does not protect online DeepSIC (excess +0.765). The jamming column is a control, not an attack these receivers are built to resist: at the same SIR, jamming spreads its power over the data symbols, 250 times the energy of the pilot attack, and raises the SER of every receiver by about the same amount (+0.813 to +0.835); Citadel-Rx rejects 13/200 of the jammed blocks, whose pilots are clean.

B. Failure of Rollback With a Single Memory

CRC-Rollback is online DeepSIC with the CRC gate G2 at a fixed slack of 0.01 ($\tau_2 = 0.99^L$), without CRC selection, and with rollback to the last accepted weights. Under every white-box and transfer attack of Table VII its excess is several times that of Citadel-Rx, and at SIR 0 and −10 dB and under transfer it does not recover after the window (+0.346, +0.637, +0.314); at +10 dB it does (+0.0017). The cause is the reference (Fig. 2). The only reference of CRC-Rollback is LMMSE on the least-squares estimate from the same pilots, and the attack degrades that estimate: on attacked blocks the reference passes a fraction 0.19 of the CRCs, against 0.71 on the same blocks of the clean twin. A poisoned fast level that is merely less bad than this reference passes the relative test: the first attacked block was accepted on 10 of 10 seeds (41/400 attacked blocks accepted in all). The accepted state then becomes the rollback state; a later update starts from the last accepted state and, in most blocks, fails the test and is reverted to it. After the attack the receiver rejects a fraction 1.00 of the clean blocks, and over the last ten blocks of the run its excess SER is still +0.351. Rollback with all three gates fails in the same way (5 seeds: 22/200 attacked blocks accepted, excess +0.316 at the end of the run), and so does Citadel-Rx with its own gates, relative slack and CRC selection but one memory ($K = 1$: excess +0.316 in the attack window; Section VI-G).

Citadel-Rx compares the fast level with the better of LMMSE and the anchor detector, which detects the data with the slow memory and does not use the pilots of the block. Under attack this reference passes 0.76 of the CRCs (0.78 on the clean twin) while the poisoned fast level passes 0.21; 3/400 attacked blocks were accepted, on 3 of the 10 seeds, and the excess over the last ten blocks is $-8.4 \cdot 10^{-5}$. The two parts of the principle depend on each other: the gate protects the memory, and it can only do so because it judges against the memory.

C. Attacks Built Against the Defence

Every attack of Table IX was built against the defence, white-box unless stated (the grey-box attacker knows the defence and its settings, not its weights; the query attacker only its NACK feedback), and run on test seeds 10–14. Against Citadel-Rx at SIR 0 dB the 13 attacks, among them the rank-4 physical attacker, which is plain A1 (see the caption), give excess SERs between +0.0171 and +0.0198, with rejection rates between 0.950 and 1. Plain A1 on the same seeds gives +0.0196 to +0.0199 (two sets of runs, on different machines). Deeper unrolling, more steps and restarts, the soft-gate objective with the thresholds and running statistics of G1 and G2 (G3 and the tightening are not modelled), expectation over noise, a Citadel-Rx surrogate, a search on the NACK feedback (one query per attacked block, so not a converged search; Table III) and a budget on the users' signal energy alone get about as few poisoned blocks accepted as plain A1 (rejection at least 0.990, against 0.990 and 0.995 for A1). So does a physical attacker with two or four antennas, also when its channel estimate is in error (at least 0.985). A physical attacker with a single antenna gets more poisoned blocks accepted (190/200 rejected) and gives the lowest excess at SIR 0 dB (+0.0171). At SIR 10 dB the excess is +0.0103 to

TABLE IX
ATTACKS BUILT AGAINST THE DEFENCE, SIR 0 DB UNLESS STATED, TEST SEEDS 10–14 (SEEDS PER ROW). [a] SOFT GATES, UNROLL 10, PGD 50, 2 RESTARTS, EOT OVER 2 NOISE DRAWS; [b] THE SAME WITHOUT SOFT GATES, WHICH ONLINE DEEPSIC DOES NOT HAVE. WITH AS MANY ANTENNAS AS THE RECEIVER, THE RANK-4 PHYSICAL ATTACKER REACHES EVERY PERTURBATION: IT IS A1, AND ITS RUNS ARE IDENTICAL TO THOSE OF THE A1 ROW OF TABLE XIII. CSI: CHANNEL STATE INFORMATION.

Attack	Receiver	Seeds	Excess SER [95% CI]	Window SER	Rej.
Unroll 10	Citadel-Rx	5	+0.0193 [0.0122, 0.0286]	0.0348	0.995
Unroll 50	Citadel-Rx	5	+0.0192 [0.0120, 0.0284]	0.0346	1.000
PGD 100, 5 restarts	Citadel-Rx	5	+0.0197 [0.0124, 0.0291]	0.0351	1.000
Soft gates, unroll 10	Citadel-Rx	5	+0.0193 [0.0121, 0.0288]	0.0348	1.000
Combined[a]	Citadel-Rx	5	+0.0197 [0.0124, 0.0291]	0.0351	1.000
Combined[b]	Online DeepSIC	5	+0.858 [0.810, 0.900]	0.891	–
Unroll 10, SIR 10 dB	Citadel-Rx	5	+0.0116 [0.0067, 0.0179]	0.0271	0.955
Soft gates, SIR 10 dB	Citadel-Rx	5	+0.0103 [0.0057, 0.0164]	0.0257	1.000
PGD 100 x 5, SIR 10 dB	Citadel-Rx	5	+0.0107 [0.0061, 0.0170]	0.0262	0.990
Rank-1 physical	Citadel-Rx	5	+0.0171 [0.0114, 0.0241]	0.0325	0.950
Rank-1 physical	Online DeepSIC	5	+0.224 [0.172, 0.297]	0.256	–
Rank-2 physical	Citadel-Rx	5	+0.0192 [0.0124, 0.0280]	0.0347	0.995
Rank-2 physical	Online DeepSIC	5	+0.581 [0.531, 0.632]	0.613	–
Rank-2, CSI error 0.1	Citadel-Rx	5	+0.0193 [0.0127, 0.0279]	0.0348	1.000
Rank-2, CSI error 0.1	Online DeepSIC	5	+0.518 [0.457, 0.580]	0.550	–
Rank-2, CSI error 0.01	Citadel-Rx	5	+0.0195 [0.0127, 0.0282]	0.0350	0.985
Rank-2, CSI error 0.01	Online DeepSIC	5	+0.577 [0.523, 0.630]	0.609	–
Rank-4 physical	Citadel-Rx	5	+0.0196 [0.0126, 0.0287]	0.0351	0.995
Rank-4 physical	Online DeepSIC	5	+0.769 [0.697, 0.834]	0.801	–
Grey-box (Citadel-Rx surrogate)	Citadel-Rx	5	+0.0198 [0.0127, 0.0292]	0.0353	0.995
Query (NACK feedback)	Citadel-Rx	5	+0.0192 [0.0120, 0.0285]	0.0346	1.000
Query (NACK feedback)	Online DeepSIC	5	+0.522 [0.449, 0.589]	0.555	–
Signal-referenced budget	Citadel-Rx	5	+0.0193 [0.0122, 0.0286]	0.0347	0.990
Signal-referenced budget	Online DeepSIC	5	+0.740 [0.658, 0.813]	0.772	–

+0.0116. The 8 of these attacks that were also run against online DeepSIC (Combined[b] in place of Combined[a]) raise its SER by +0.224 (rank-1 physical attacker) to +0.858. All used their full budget, and with five seeds their excess is not distinguishable from plain A1; what the table shows is that none of these attacks brings Citadel-Rx near the excess of the undefended receivers.

When the gates reject nearly every attacked block, the excess depends on the power of the pilot perturbation (SIR 0 against 10 dB), not on how it was optimised: it is the price of rejecting a block with poisoned pilots; an attack that also reaches the data and gets more blocks accepted costs more (Section VI-D). The data of the window are detected with the anchor, which has not adapted to the block, and the LMMSE alternative of the CRC selection uses the poisoned channel estimate (its pass fraction falls from 0.71 on the clean twin to 0.20 on attacked blocks). On the poisoned blocks at the headline cell, the verdict of G2 on its own is below one half for 99% of them, that of G1 for 88% and that of G3 for 14%: the decision-level gates G1 and G2 account for the rejections, and G3 flags few poisoned blocks at its threshold.

TABLE X
JOINT PILOT AND DATA ATTACK ON THE REFERENCES (A4), WHITE-BOX SIR 0 DB, 6 PGD STEPS OVER ALL DATA SUB-BLOCKS, TEST SEEDS 10–14 (ROWS MARKED 15–19: SEEDS 15–19; ROWS MARKED 1ST: SEEDS 10 AND 11), WITH AND WITHOUT THE ABSTAIN STATE (FIRST VERSION: REJECTS EVERY BLOCK OF ITS CLEAN RUNS AS WELL, SEE TEXT); ONE ROW WITH 20 PGD STEPS; JAMMING CONTROLS WITH THE DATA POWER OF A4.

Attack	State	Seeds	Excess SER [95% CI]	Window SER	Rej.
$\alpha = 0.25$	–	5	+0.0337 [0.0204, 0.0470]	0.0492	0.760
$\alpha = 0.25$	abstain, 1st	2	+0.0102 [0.0074, 0.0133]	0.0321	1.000
$\alpha = 0.25$	abstain	5	+0.0213 [0.0154, 0.0277]	0.0368	0.960
$\alpha = 0.25$	abstain, 15–19	5	+0.0214 [0.0163, 0.0265]	0.0364	0.955
$\alpha = 0.25$	20 PGD steps	5	+0.0478 [0.0279, 0.0689]	0.0633	0.775
jamming	0.75 of power	5	+0.785	–	–
$\alpha = 0.75$	–	5	+0.0276 [0.0194, 0.0368]	0.0431	0.985
$\alpha = 0.75$	abstain, 1st	2	+0.0126 [0.0084, 0.0180]	0.0344	1.000
$\alpha = 0.75$	abstain	5	+0.0249 [0.0169, 0.0342]	0.0404	1.000
$\alpha = 0.75$	abstain, 15–19	5	+0.0242 [0.0175, 0.0312]	0.0392	1.000
jamming	0.25 of power	5	+0.481	–	–

TABLE XI
ABRUPT CHANNEL REDRAW AT BLOCK 50, WITHOUT AND WITH THE DENIAL-OF-ADAPTATION SPOOF ON BLOCKS 40–89 (SIR 10 DB), AND A 6 DB SNR DROP AT BLOCK 50; TEST SEEDS 10–14. SER AFTER THE CHANGE (NO ATTACK), FALSE REJECTION AFTER IT, EXCESS SER OF THE SPOOF IN AND AFTER ITS WINDOW, SER AFTER THE SNR DROP.

Receiver	Seeds	SER after redraw	False rej.	Spoof excess	After spoof	SER after drop
Citadel-Rx	5	0.250	0.180	+0.0968	+0.0096	0.457
+ streak recovery	5	0.250	0.180	+0.0967	+0.0094	0.457
Online DeepSIC	5	0.224	–	+0.168	+0.0432	0.468
LMMSE	5	0.351	–	+0.204	0	0.479

D. Attacks on the References and on Adaptation

A4 splits the attacker's power between the pilots and the data so that the references the gates compare against degrade together with the fast level (Table X). At the headline setting and without a channel redraw, it gives Citadel-Rx the largest excess SER of the attacks on its pilots and references; jamming the data (a control), attacks during a channel redraw (Tables XI and XII) and the settings where Citadel-Rx fails (16-QAM and the coded scene, Sections VI-H and VI-J) give more. With a quarter of the power on the pilots, the reference passes 0.65 of the CRCs on attacked blocks (0.78 on the clean twin), close enough to the poisoned fast level (0.46) that only 152/200 attacked blocks are rejected, and the excess SER is +0.0337, against +0.0196 to +0.0199 for A1 on the same seeds; with three quarters on the pilots it is +0.0276. More optimisation raises the excess, with overlapping intervals: with 20 PGD steps instead of 6 the excess at $\alpha = 0.25$ is +0.0478 and 155/200 attacked blocks are rejected, against +0.0337 and 152/200 with 6 steps, so the 6-step rows may understate this attacker. Its surrogate models G1 against LMMSE only and

TABLE XII
DENIAL OF ADAPTATION AS INTENDED: WHITE-BOX A1 AT SIR 0 DB ON BLOCKS 40–89 WHILE THE CHANNEL IS REDRAWN AT BLOCK 50, TEST SEEDS 10–14. REJECTION OF THE ATTACKED BLOCKS, SER AFTER THE REDRAW WITHOUT ATTACK, EXCESS SER IN AND AFTER THE ATTACK, AND STREAK-RECOVERY RE-FITS TRIED/KEPT. THE RUNS WITHOUT ATTACK ARE NOT THOSE OF TABLE XI: EACH RAN ON THE MACHINE OF ITS ATTACKED RUN, AND ON SEEDS 10 AND 11 THAT MACHINE DIFFERS BETWEEN THE TWO TABLES (SER AFTER THE REDRAW FOR CITADEL-RX 0.251 HERE, 0.250 THERE).

Receiver	Seeds	Rej.	SER after redraw	Excess in window	Excess after	Re-fits tried/kept
Citadel-Rx	5	0.988	0.251	$+0.0543$	$+0.0511$	–
+ streak recovery	5	0.888	0.251	$+0.0461$	$+0.0462$	51/25
Online DeepSIC	5	–	0.224	$+0.632$	$+0.0985$	–
LMMSE	5	–	0.351	$+0.342$	0	–

TABLE XIII
PER-USER GATING AGAINST AN ATTACK ON ONE USER'S PILOTS (A2, USER 1) AND AGAINST A1, WHITE-BOX SIR 0 DB, TEST SEEDS 10–14: EXCESS SER OF ALL USERS AND OF USER 1; REJECTION OF THE ATTACKED BLOCKS (BLOCK GATE) OR REVERTS OF USER 1 / OF THE OTHER USERS (PER-USER GATE).

Attack	Receiver	Seeds	Excess SER [95% CI]	Excess SER user 1	Rej. or reverts
A2 (user 1)	Citadel-Rx	5	$+0.0193$ [0.0121, 0.0286]	$+0.0040$	0.995
A2 (user 1)	+ per-user gating	5	$+0.0212$ [0.0143, 0.0297]	$+0.0044$	0.960 / 0.970
A2 (user 1)	Online DeepSIC	5	$+0.468$ [0.408, 0.528]	$+0.307$	–
A1	Citadel-Rx	5	$+0.0196$ [0.0126, 0.0287]	$+0.0042$	0.995
A1	+ per-user gating	5	$+0.0214$ [0.0141, 0.0301]	$+0.0045$	0.965 / 0.970

G2, with untightened thresholds, and omits G3. The A4 rows therefore describe this attacker, not every joint attacker; more than 20 PGD steps and an attacker with an exact model of the gates were not run. An attacker that reaches the data does far more harm by jamming them: the same data power as noise raises the SER by $+0.785$ and $+0.481$ for $\alpha = 0.25$ and 0.75.

The abstain state is meant to answer A4. Its first version did not work: its running statistics were updated only on accepted blocks and its variance floor was binomial over thousands of CRC checks, so the fading of the channel alone, which lowers the reference pass fraction from 0.987 to 0.963, tripped it at block 4–5, and from then on it rejected every block, clean or attacked (191/200 blocks of its clean twins; the lower excess of these rows in Table X is that of a receiver that no longer adapts in either run). The final version (Section III) does not trip on clean runs without a channel redraw: it abstained on 0/475 clean blocks of seeds 10–14, on 0/475 of the fresh seeds 15–19 and on 0/250 of the blocks after a 6 dB SNR drop (after the drop no slot is verified for every user, so there the statistic is that of the LMMSE decisions). It abstains on every attacked block of A1 and of A4 at $\alpha = 0.75$, whose energy on the pilots inflates the noise estimate, and on 96% of those of A4 at $\alpha = 0.25$ on seeds 10–14 (95% on seeds 15–19). It lowers the excess of A4, with overlapping intervals, to near that of A1 with the same state: $+0.0213$ and $+0.0249$ on seeds 10–14, against $+0.0337$ and $+0.0276$ without it and $+0.0192$ for A1 (on seeds 15–19, $+0.0214$ and $+0.0242$, against $+0.0171$ for A1). After a legitimate redraw it abstains on 24.4 of the 50 remaining blocks on average (5 seeds) and on all of them on 2: when the CRCs of one user fail after the redraw, no symbol slot is verified for every user (the verified fraction falls from 0.81 to 0.11), the statistic falls back to the residual of the LMMSE decisions, and that residual follows their errors. On the seeds where it abstains on every remaining block, the re-base after 50 consecutive abstentions falls on the last block of the run, so these runs do not test it. With the final abstain state the SER over the blocks after the redraw is 0.278, against 0.250 without it (Table XI; the two sets of runs are not paired). The final version is therefore not a usable defence either; a residual on the verified symbols of each user is the next step.

An abrupt channel redraw at block 50 is a legitimate change that the gates must not suppress (Table XI). After it Citadel-Rx rejects a fraction 0.180 of the clean blocks and its SER over the remaining blocks is 0.250, against 0.224 for online DeepSIC and 0.351 for LMMSE: the gates slow the re-adaptation, a cost of the defence. After a 6 dB SNR drop, which the receivers are not told about, all three are close (0.457, 0.468, 0.479). The denial spoof did not achieve its aim of tripping the gates on every block: they rejected fractions between 0.04 and 0.32 of the spoofed blocks on the 5 seeds. It acted instead as an accepted perturbation during the change, raising the SER in its window by $+0.0968$ for Citadel-Rx, $+0.168$ for online DeepSIC and $+0.204$ for LMMSE; in the ten blocks after it, the excess was $+0.0096$ for Citadel-Rx and $+0.0432$ for online DeepSIC. Streak recovery was attempted 22 times in these runs and kept 2 times, changing 2 of 15 runs.

Denial as intended needs an attack that the gates reject on nearly every block. Strong poisoning does that: A1 at SIR 0 dB on blocks 40–89, while the channel is redrawn at block 50 (Table XII), is rejected on 0.988 of the attacked blocks, so Citadel-Rx does not adapt to the new channel while the attack lasts. It pays $+0.0543$ in excess SER in the window and still $+0.0511$ in the ten blocks after it, against $+0.632$ in the window for online DeepSIC, which adapts to the poisoned pilots, and $+0.342$ for LMMSE. Streak recovery is meant for this case: it re-fitted from the anchor on CRC-verified data 51 times and kept 25 of these re-fits, and the excess was $+0.0461$ in the window and $+0.0462$ after it, with 0.888 of the attacked blocks rejected. Table XII has no intervals, so with 5 seeds the reduction is a point estimate, and most of the cost remains. White-box attacks at SIR 10 dB are also rejected on nearly every block when the channel is not redrawn (Table IX), so denial may need less power than this; the weakest perturbation that the gates still reject, which sets the attacker's cost of denial, was not searched for, and denial was not run during an SNR drop.

An attack on one user. Per-user gating is meant for an attacker who poisons the pilots of one user only (A2, Table XIII). It does not help here: the excess SER is $+0.0212$ with it and $+0.0193$ without. A2 perturbs the received pilots, which the networks of all users take as input. The block-level gates G1 and G3 reject the whole block on 0.835 of the attacked blocks, and on most of the others the G2 test of the other users also fails, so the other users are reverted on 0.970 of the attacked blocks and the attacked user on 0.960.

TABLE XIV
RETENTION OF THE SLOWEST LEVEL: PAIRED DISPLACEMENT D_K AGAINST THE STRUCTURAL BOUND (8) AND THE DESIGN BOUND (7), EACH THE LARGEST VALUE IN THE WINDOW, AVERAGED OVER SEEDS; IN EVERY RUN D_K IS BELOW BOTH BOUNDS AT EVERY BLOCK OF THE WINDOW. A1: PAIRED RUNS WITH LEVEL SNAPSHOTS, FIXED WINDOW 30–69, SEEDS 10–14. A3*: BLOCKS 30–329 OF 400 (900 BLOCKS: 30–929 OF 1000), PERTURBATION NORM RELATIVE TO THE RECEIVED PILOTS, SEEDS 10–11; WITH TWO SEEDS AN INTERVAL REFLECTS MOSTLY THE VARIATION BETWEEN BLOCKS, NOT BETWEEN SEEDS.

Attack	Seeds	Excess SER [95% CI]	Rej.	D_K	Struct.	Design
A1, SIR 0 dB	5	+0.0137 [0.0098, 0.0180]	0.995	4.17	15.7	18.0
A1, SIR 10 dB	5	+0.0073 [0.0048, 0.0099]	0.945	3.73	13.3	18.0
A3*, 5%	2	+0.0017 [0.0014, 0.0021]	0.002	8.48	11.3	90.0
A3*, 20%	2	+0.0122 [0.0099, 0.0147]	0.225	13.30	19.5	90.0
A3*, 50%	2	+0.0109 [0.0086, 0.0136]	0.998	22.03	57.3	90.0
A3*, 50%, guard	2	+0.0110 [0.0087, 0.0137]	0.998	18.72	56.2	90.0
A3*, 5%, guard	2	+0.0017 [0.0014, 0.0020]	0.003	6.86	10.6	90.0
A3*, 5%, 900 blocks	2	+0.0016 [0.0014, 0.0018]	0.001	14.94	16.7	261.0

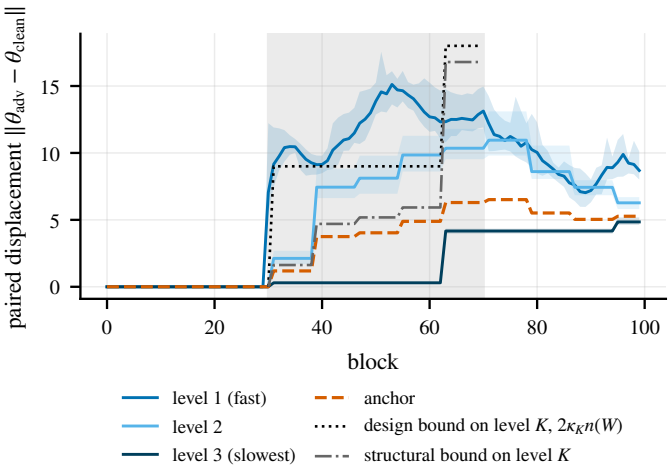


Fig. 3. Paired displacement of every level and of the anchor under A1 at SIR 0 dB, window 30–69 shaded (mean over seeds 10–14; for the levels, band from the smallest to the largest seed). The design bound (7) and the structural bound (8) apply to the slowest level only; the structural bound is drawn as the largest over the seeds, Table XIV gives its mean.

On clean blocks 0.221 of the user updates are reverted. Under A1, which poisons all users, per-user gating gives +0.0214, against +0.0196 without it.

E. Retention Under a Slow Small-Budget Attack

Table XIV and Fig. 3 give the paired displacement for the A1 runs with level snapshots. The largest displacement of the slowest level in the window is 4.17 at SIR 0 dB and 3.73 at 10 dB, below the structural bound (15.7 and 13.3) and the design bound (18.0), all means over seeds of the largest value in the window; on every seed D_K stays below both bounds at every block of the window. After the window D_K rises once more, at the next consolidation of the slowest level (Fig. 3), to 4.85 at SIR 0 dB and 5.13 at 10 dB (means over seeds of the largest value in the run), below the design bound with the consolidations counted to the end of the run (27.0); the table and the bounds drawn cover the window only. The anchor,

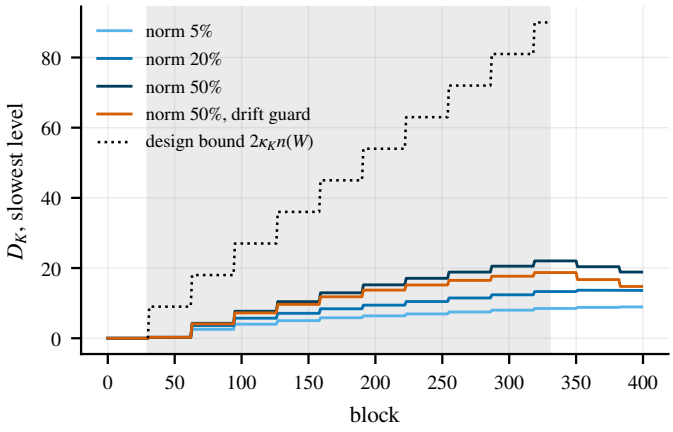


Fig. 4. A3* on blocks 30–329 (shaded): paired displacement of the slowest level for each perturbation size, and at 50% also with the drift guard (mean over seeds 10–11, band from the smaller to the larger seed), and the design bound (7); the structural bounds (8) are in Table XIV.

with which rejected blocks are detected, is displaced by up to 6.29 in the window (mean over seeds of the largest value). The displacement of the slowest level is not a footprint the attack left in the memory. Over the window the slowest level of the clean run moved by 8.4 and that of the attacked run by 4.8, in nearly the same direction (cosine 0.94 between the two movements). Rejected blocks enter the consolidation with weight $\gamma_b \approx 0$, so the attacked memory follows the channel more slowly: it moved 57% as far as the clean memory at SIR 0 dB and 79% at 10 dB, and D_K measures that lag.

A3* spreads its perturbation over 300 of 400 blocks, each optimised for the soft-gate objective so that it is accepted (Table XIV, Fig. 4). With a norm of 5% of that of the received pilots (SIR 26 dB) it succeeds: 1/600 attacked blocks are rejected. This is the case the retention bounds are for, but not their worst case: A3* optimises each block on its own for an accepted data loss, and its objective has no term for the displacement of the slow level and no target common to its blocks. An attacker that maximises D_K subject to acceptance was not run. Over the window the slowest level of the attacked run moved as far as that of the clean run (27.0 and 26.8), in nearly the same direction (cosine 0.95); their paired displacement reached 8.48, below the structural bound (11.3) and far below the design bound (90.0). The accepted perturbation cost +0.0017 in SER during the attack and $+1.7 \cdot 10^{-4}$ in the 30 blocks after it. At 20% the gates reject 135/600 attacked blocks (excess +0.0122, $D_K = 13.30$ against a structural bound of 19.5); at 50% they reject 599/600, the attacked memory nearly stands still (it moved 23% as far as the clean one), and $D_K = 22.03$ is again a lag, not a push. In the 30 blocks after the attack the excess is at most +0.0018 for every A3* setting. Over 300 blocks the design bound is loose, while the structural bound is 1.3 to 3.0 times the measured displacement. At 50%, where nearly every attacked block is rejected, the drift guard lowered D_K to 18.72 by pulling both runs towards the pre-trained weights (the clean memory moved 22.0 instead of 26.8), at an excess of +0.0110. Against the accepted 5% perturbation, the case it is meant for, the perturbation was still accepted (2/600 attacked

TABLE XV
FURTHER BASELINES UNDER WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–14. DD: DECISION-DIRECTED.

Receiver	Seeds	Clean SER	Excess SER [95% CI]	Window SER
LMMSE	5	0.0562	$+0.319$ [0.275, 0.362]	0.372
Robust LMMSE (Huber)	5	0.0562	$+0.268$ [0.226, 0.306]	0.321
Rollback, three gates	5	0.0330	$+0.237$ [0.194, 0.283]	0.269
L2-SP, weight 0.01	5	0.0380	$+0.762$ [0.705, 0.814]	0.799
EMA teacher	5	0.0288	$+0.438$ [0.329, 0.546]	0.466
EWC	5	0.0349	$+0.772$ [0.720, 0.819]	0.806
CRC-verified DD	5	0.0259	$+0.752$ [0.671, 0.827]	0.777
CoTTA-style	5	0.0263	$+0.245$ [0.192, 0.299]	0.271
EATA-style	5	0.0349	$+0.772$ [0.720, 0.819]	0.806
Drift detection	5	0.0491	$+0.686$ [0.617, 0.748]	0.735
Single memory (Titans)	5	0.0544	$+0.732$ [0.703, 0.754]	0.784
Median-of-means gradient	5	0.0300	$+0.648$ [0.563, 0.726]	0.677
L2-SP, weight 0.001	5	0.0366	$+0.785$ [0.728, 0.838]	0.821
L2-SP, weight 0.1	5	0.0387	$+0.590$ [0.551, 0.629]	0.628
L2-SP, weight 1	5	0.0395	$+0.136$ [0.109, 0.161]	0.175

blocks rejected), both memories moved less (22.0 instead of 27.0) and D_K fell from 8.48 to 6.86 (structural bound 10.6), but the excess SER did not fall ($+0.0017$ against $+0.0017$ without the guard, $+3.5 \cdot 10^{-4}$ after the window): the guard shrinks what an accepted perturbation leaves in the memory without lowering its cost. Over 900 attacked blocks of 1000 (A3* at 5%, 2 seeds) the perturbation was again accepted (1/1800 attacked blocks rejected), D_K reached 14.94, below the structural bound (16.7) and the design bound (261.0), at an excess of $+0.0016$ in the window and $-3.2 \cdot 10^{-4}$ after it.

Without attack the SER does not drift over a long run. Over 2000 clean blocks of the SED channel with cosine fading (2 seeds) the SER of Citadel-Rx was 0.0149 over blocks 5–499 and 0.0149 over the last 500, with 1/3990 blocks falsely rejected; that of tuned online DeepSIC was 0.0321 and 0.0306.

F. Other Baselines

None of the further baselines resists A1 (Table XV); at their default settings their excess SERs lie between $+0.237$ and $+0.772$. Robust statistics give little protection against a perturbation spread over all pilots of a block: Huber-robust LMMSE has $+0.268$ against $+0.319$ for LMMSE, and the median-of-means gradient $+0.648$ against $+0.768$ for online DeepSIC (Table VII, ten seeds). Anchoring to the pre-trained weights buys protection at the cost of adaptation: L2-SP limits the damage more as its weight grows, from $+0.785$, $+0.762$ and $+0.590$ at weights 0.001, 0.01 (the default) and 0.1 down to $+0.136$ at weight 1, the smallest excess of any baseline, with a clean SER of 0.0395. The sweep stops at weight 1; its limit, the pre-trained DeepSIC without test-time training, has zero excess by construction and was not run, so what adaptation gains over it on clean blocks is not measured. The EATA-style filter never fired, because the attacked receiver is confidently wrong: its mean data entropy stayed at or below 0.38 nats against a threshold of 0.55 nats, so the EATA-style receiver coincides with EWC. CRC-verified decision-directed training still adapts on the poisoned pilots ($+0.752$); the CoTTA-style teacher with stochastic restore keeps $+0.245$, and the single-memory receiver with the fast learner of Citadel-Rx and no gates $+0.732$, so the fast learner is not what protects Citadel-Rx. Without attack, none of the baselines of Table XV closes most of the clean gap between tuned online DeepSIC (0.0330, Table VII) and Citadel-Rx (0.0158): the lowest clean SERs are 0.0259 for CRC-verified decision-directed training and 0.0263 for the CoTTA-style receiver, and L2-SP and EWC raise the clean SER. Citadel-Rx without the CRC selection has 0.0317 (Table XVI), above four of the baselines: its clean gain comes with the selection, to which the slow memory contributes the anchor detection as a candidate, not from the memory or the anchor alone.

TABLE XVI
ABLATIONS UNDER WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–14.

Configuration	Seeds	Clean SER	Excess SER [95% CI]	Rej.	False rej.
Complete receiver	5	0.0158	$+0.0199$ [0.0127, 0.0291]	0.990	0.002
$K = 1$ (gates, rollback, no slow level)	5	0.0340	$+0.316$ [0.250, 0.396]	0.960	0.086
$K = 2$, $C = (1, 8)$	5	0.0170	$+0.0242$ [0.0181, 0.0306]	0.985	0.000
No gates	5	0.0158	$+0.204$ [0.148, 0.265]	0.000	0.000
G1 only	5	0.0158	$+0.0218$ [0.0145, 0.0309]	0.970	0.000
G2 only	5	0.0157	$+0.0194$ [0.0124, 0.0284]	0.995	0.000
G3 only	5	0.0158	$+0.174$ [0.131, 0.218]	0.160	0.000
No anchor ($\lambda = 0$)	5	0.0228	$+0.0130$ [0.0077, 0.0200]	1.000	0.133
No fallback	5	0.0158	$+0.0859$ [0.0553, 0.122]	0.995	0.000
No CRC selection	5	0.0317	$+0.0053$ [$-1.3 \cdot 10^{-4}$, 0.0127]	0.995	0.000
No consolidation clip	5	0.0158	$+0.0194$ [0.0123, 0.0285]	0.995	0.000

G. Ablations

Table XVI removes one part at a time; on the same seeds the complete receiver has an excess of $+0.0199$ [0.0127, 0.0291] and a clean SER of 0.0158. Without the slow levels ($K = 1$, gates, rollback and fallback to the last accepted weights kept) the excess is $+0.316$: the receiver fails like CRC-Rollback, for the reason of Section VI-B, and it also rejects 0.086 of the clean blocks. Two levels, $C = (1, 8)$, give $+0.0242$, not distinguishable from three levels with five seeds. Without the fallback, a rejected block is still reverted but its data are detected with the rejected update: $+0.0859$. Without gates the excess is $+0.204$. With either decision-level gate alone the excess is not distinguishable from that of the complete receiver with five seeds (G1 $+0.0218$, G2 $+0.0194$); the channel-prior gate G3 does not ($+0.174$, rejecting 0.160 of the poisoned blocks). Removing the proximal pull to the anchor lowers the excess to $+0.0130$ only because the clean twin is worse (0.0228) and rejects 0.133 of its blocks. Without the CRC selection the clean SER doubles (0.0317) and the excess nearly vanishes ($+0.0053$): the selection is most of what the receiver gives up while it rejects. The clip was active in 5/150 consolidations of the clean twins and 12/150 of the attacked runs at the headline cell, and without it the excess ($+0.0194$) is not distinguishable with five seeds; it provides the guarantee of Theorem 1(i), not the measured robustness. Three parts kept from the first version are not ablated here, so their necessity is not established: the reversion of a rejected update (the row without fallback still reverts), the delta-rule

TABLE XVII
BLOCK LENGTH, 4×16 AND 8×32 ARRAYS (STRETCHED SED PROFILE), 16-QAM AND TWO CHEAPER VARIANTS, WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–14 (SEEDS PER ROW); WALL TIME PER BLOCK OF SINGLE-THREADED RUNS ON THE LAPTOP.

Setting	Receiver	Seeds	Clean	Excess SER [95% CI]	Rej.	s/ block
$N_d = 1000$, $N_p = 32$	Citadel-Rx	5	0.0213	$+0.0206$ [0.0141, 0.0282]	0.980	1.3
$N_d = 1000$, $N_p = 32$	Online Deep-SIC	5	0.0387	$+0.671$ [0.569, 0.757]	–	0.1
$N_d = 1000$, $N_p = 100$	Citadel-Rx	5	0.0160	$+0.0189$ [0.0123, 0.0271]	0.995	1.6
$N_d = 1000$, $N_p = 100$	Online Deep-SIC	5	0.0334	$+0.763$ [0.692, 0.828]	–	0.2
$N_d = 5000$, $N_p = 32$	Citadel-Rx	5	0.0202	$+0.0215$ [0.0142, 0.0310]	0.965	1.5
$N_d = 5000$, $N_p = 32$	Online Deep-SIC	5	0.0365	$+0.687$ [0.600, 0.760]	–	0.2
$N_d = 5000$, $N_p = 100$	Citadel-Rx	5	0.0160	$+0.0200$ [0.0125, 0.0299]	0.995	1.9
$N_d = 5000$, $N_p = 100$	Online Deep-SIC	5	0.0333	$+0.774$ [0.705, 0.838]	–	0.2
$U \times N = 4 \times 16$	Citadel-Rx	5	$6.8 \cdot 10^{-4}$	$+0.0040$ [0.0020, 0.0066]	0.910	–
$U \times N = 4 \times 16$	Online Deep-SIC	5	0.0037	$+0.818$ [0.738, 0.889]	–	–
$U \times N = 4 \times 16$	LMMSE	5	0.0090	$+0.0812$ [0.0507, 0.114]	–	–
$U \times N = 8 \times 32$	Citadel-Rx	5	0.0023	$+0.0095$ [0.0063, 0.0141]	0.995	–
$U \times N = 8 \times 32$	Online Deep-SIC	5	0.0151	$+0.926$ [0.886, 0.958]	–	–
$U \times N = 8 \times 32$	LMMSE	5	0.0253	$+0.539$ [0.498, 0.570]	–	–
16-QAM, 20 dB	Citadel-Rx	5	0.120	$+0.558$ [0.478, 0.640]	0.700	5.5
16-QAM, 20 dB	Online Deep-SIC	5	0.0512	$+0.932$ [0.914, 0.948]	–	0.5
16-QAM, 20 dB	LMMSE	5	0.0094	$+0.951$ [0.947, 0.954]	–	0.1
30 fast steps	Citadel-Rx	5	0.0211	$+0.0190$ [0.0130, 0.0266]	0.990	1.9
last layer adaptable	Citadel-Rx	5	0.0191	$+0.0198$ [0.0128, 0.0290]	0.995	2.6

decay (Section VI-K varies δ by a factor of two only) and the surprise-modulated step size.

H. Block Length, Arrays, Modulation and Cost

Shorter blocks do not break the defence (Table XVII): with $N_d \in \{1000, 5000\}$ and $N_p \in \{32, 100\}$ the excess of Citadel-Rx stays between $+0.0189$ and $+0.0215$, with at least 0.965 of the poisoned blocks rejected, against $+0.671$ to $+0.774$ for online DeepSIC. Fewer CRC sub-blocks make the pass fractions noisier: with $N_d = 1000$ the clean false rejection is above the 2% cap of the calibration rule, 0.021 (10/475 blocks) at $N_p = 100$ and 0.025 (12/475) at $N_p = 32$, against 0.004 and 0.006 with $N_d = 5000$. The floor of the relative slack was set for $n = 781$ sub-blocks per user and does not follow the block length; the derived slack, which does, was not run at these block lengths. The SED profile $e^{-|i-j|}$ puts 99% of the channel energy of an 8×32 array on 9 antennas, so the larger arrays use it stretched over the array, $e^{-|iU/N-j|}$, which keeps the structure of the interference and spreads the energy over 31 of 32 antennas. The stretched profile was adopted after test runs with the unstretched one (5 seeds at 8×32, 2 at 4×16), which are not in Table XVII. In those runs Citadel-Rx had an excess of $+0.0407$ (clean SER 0.0601, 199/200 poisoned blocks rejected), online DeepSIC $+0.841$ and LMMSE $+0.442$ at 8×32, and $+0.0142$, $+0.772$ and $+0.164$ at 4×16. With the stretched profile and 4×16 antennas the excess of Citadel-Rx is $+0.0040$, with 0.910 of the poisoned blocks rejected, against $+0.818$ for online DeepSIC and $+0.0812$ for LMMSE; with eight users and 8×32 antennas it is $+0.0095$ (0.995 rejected), against $+0.926$ and $+0.539$, with no clean block falsely rejected in either array (0/475 and 0/475).

Citadel-Rx fails at 16-QAM and 20 dB. There the pre-trained network passes a fraction 0.11 of the CRCs on clean blocks and the LMMSE reference 0.93, so the relative gates reject 98% of the clean blocks, the network hardly ever adapts, and the clean SER (0.120) is worse than that of online DeepSIC (0.0512) and of LMMSE (0.0094). The network is also narrower here than the width rule would give (256 hidden units, against $64 \cdot 16 = 1024$), which may contribute to its weak clean pass fraction; a wider network was not tested. The CRC selection keeps the gate's choice in the sub-blocks where nothing passes, so it does not recover the better LMMSE decisions. Under attack Citadel-Rx is not protected either: its excess SER, $+0.558$ [0.478, 0.640], is below those of online DeepSIC ($+0.932$) and LMMSE ($+0.951$) but far above the $+0.0188$ of the headline cell (QPSK, 10 dB). The gates reject a smaller fraction of the poisoned blocks (0.700) than of the clean ones (0.979), because the attack also degrades the LMMSE reference against which the relative gates judge the network, and the gates loosen with it. The relative gates presuppose a network that is at least as good as the model-based reference on clean blocks; when it is not, a persistent clean false rejection is the symptom, and falling back to the reference is the obvious remedy, not evaluated here.

The two cheaper variants keep the protection: 30 fast steps per block give an excess of $+0.0190$ at 1.93 s per block, but this variant rejects 0.152 of the clean blocks and raises the clean SER to 0.0211 (full receiver: 0.0158), and the last-layer variant (clean SER 0.0191) gives $+0.0198$ at 2.65 s. The full receiver takes 3.03 s per block, against 0.56 s for tuned online DeepSIC, 16.33 s for Meta-DeepSIC and 0.03 s for LMMSE: it takes 300 full-batch steps per stage and block where tuned online DeepSIC takes 30, and detects the data with both the fast level and the anchor. Counted in multiply-accumulates from the network dimensions (training as three times the forward pass), a block costs Citadel-Rx 9.0 giga multiply-accumulates (GMAC), 64% of it in training, against 2.2 GMAC for online DeepSIC and CRC-Rollback (a factor of 4.1 lower) and 3.8 GMAC for the variant with 30 fast steps.

I. A Second Backbone

With the joint-symbol detector in place of DeepSIC (Table XVIII), the levels, gates and fallback of Citadel-Rx keep

TABLE XVIII
SECOND BACKBONE: A JOINT-SYMBOL DETECTOR (ONE HIDDEN LAYER OF 512 UNITS, 256 CLASSES) IN PLACE OF DEEPSIC, WHITE-BOX A1 AT SIR 0 DB, TEST SEEDS 10–14.

Receiver	Seeds	Clean SER	Excess SER [95% CI]	Rej.	False rej.
Citadel-Rx	5	0.0255	$+0.0052$ [0.0035, 0.0074]	1.000	0.217
Online	5	0.0866	$+0.349$ [0.300, 0.395]	–	–
CRC-Rollback	5	0.0812	$+0.151$ [0.115, 0.190]	0.695	0.295

TABLE XIX
NR LDPC CODE BLOCKS OVER TR 38.901 UMI DROPS WITH THE ATTACKER IN THE SCENE (ONE ANTENNA, A USER'S MAXIMUM POWER, WHITE-BOX), TEST SEEDS 10–14; THE SIR FOLLOWS FROM THE GEOMETRY (MEDIAN OVER THE DROPS). BLER: CODE-BLOCK ERROR RATE AFTER DECODING.

Attacker	Receiver	Seeds	Clean BLER	Excess BLER [95% CI]	Excess SER	Rej.
35 m (-26 dB)	Citadel-Rx	5	0.0650	$+0.675$ [0.575, 0.760]	$+0.574$	0.920
	Online	5	0.169	$+0.423$ [0.316, 0.514]	$+0.414$	–
	CRC-Rollback	5	0.859	$+0.0727$ [0, 0.176]	$+0.0394$	0.960
	LMMSE	5	0.0686	$+0.779$ [0.726, 0.826]	$+0.736$	–
75 m (-3 dB)	Citadel-Rx	5	0.0650	$+0.432$ [0.271, 0.585]	$+0.497$	0.990
	Online	5	0.169	$+0.360$ [0.229, 0.493]	$+0.290$	–
	CRC-Rollback	5	0.859	$+0.0261$ [0, 0.0914]	$+0.0263$	0.990
	LMMSE	5	0.0686	$+0.477$ [0.265, 0.660]	$+0.497$	–
150 m ($+2$ dB)	Citadel-Rx	5	0.0650	$+0.243$ [0.125, 0.352]	$+0.386$	0.995
	Online	5	0.169	$+0.136$ [0.0873, 0.187]	$+0.131$	–
	CRC-Rollback	5	0.859	$+0.0287$ [0, 0.0995]	$+0.0281$	0.990
	LMMSE	5	0.0686	$+0.230$ [0.102, 0.360]	$+0.306$	–

their frozen settings, and online training is tuned on the calibration seeds (30 epochs, learning rate 0.001, clean SER 0.0828). The ordering of Table VII holds: under white-box A1 at SIR 0 dB the excess SER of Citadel-Rx is $+0.0052$ (95% interval [0.0035, 0.0074]), with 200/200 poisoned blocks rejected, against $+0.349$ for online training and $+0.151$ for CRC-Rollback, which rejects 139/200 of them. The joint detector is the weaker network on this channel (clean SER 0.0866 with online training), and the price is paid on clean blocks: Citadel-Rx rejects 103/475 (0.217) of them, against 0.002 with DeepSIC on the same seeds (Table XVI), while its clean SER, 0.0255, stays below that of online training on the same backbone. The levels and gates do not depend on DeepSIC's structure, but how often they let the network adapt depends on how good the network is.

J. A Coded NR Link in a 3GPP Scene

Each test seed is one drop of the TR 38.901 urban-micro street-canyon model [43], generated with Sionna [59] (Table V), which implements the parameter tables of V19.2 of the report: path loss, line-of-sight probability and shadow fading of the standard, a narrowband channel per 1 ms slot, the users moving at 3 km/h. Slow power control aims every user at the mean received power of a cell-edge user (150 m, non-line-of-sight, no shadowing); a user that cannot reach it at maximum power arrives weaker. The attacker is one static antenna at 35, 75 or 150 m from the base station, transmitting at a user's maximum power, with its channel from the same drop, so its SIR is not a setting: over the five drops it ranges from -34 to -9 dB at 35 m (median -26), from -18 to $+4$ dB at 75 m (median -3) and from -5 to $+14$ dB at 150 m (median $+2$). It knows the receiver (white-box) and designs its pilot perturbation through its own channel. Powers are relative to a user's maximum power: an attacker with x dB more transmit power lowers each of these SIRs by x dB ($x = 7$ to 17 for a jammer of 30–40 dBm against a user limit of 23 dBm); such an attacker was not run. The data of every user are independent NR LDPC code blocks (base graph 2, $k = 520$, $n = 1000$), each with a CRC-24A [54] (a simplification: NR attaches one CRC to the transport block, CRC-16 up to 3824 bits and CRC-24A above, and a CRC-24B to every code block only when it segments it), scrambled with the user's NR PUSCH sequence ($c_{\mathrm{init}} = n_{\mathrm{RNTI}} 2^{15} + n_{\mathrm{ID}}$) [55], decoded with 20 belief-propagation iterations from the hard symbol decisions, and the CRC after decoding gives the block error rate (BLER). The scrambling matters for the gates: the all-zero codeword has a zero CRC, so without scrambling a detector stuck on the symbol whose bits are all zero would pass every check (a first version of these runs had this flaw and is not reported). All receivers keep their frozen settings; the CRC gate uses the virtual pass fraction of Section III-D.

The defence fails in the coded scene (Table XIX). Its channel is the first channel of this paper whose phases rotate from block to block, and the networks, pre-trained on the first block, do not follow it: over blocks 5–99 the clean BLER of online DeepSIC is 0.169, against 0.0686 for LMMSE, which re-estimates the channel on every block. The phase rotation is the likely cause, not an established one: no run with the phases held fixed separates it from the other differences of this scene (mobility, path loss and shadowing, power control), and every receiver keeps the settings chosen on the SED channel. CRC-Rollback, which reverts every update that fails its test, ends at 1.000 over the last 20 blocks, in which LMMSE decodes every code block. For Citadel-Rx this is the regime of 16-QAM (Section VI-H): on clean blocks its fast level passes a virtual fraction 0.03 of the checks against 0.60 for the reference, so 453/475 clean blocks are rejected and the receiver runs on the anchor and on LMMSE through the CRC selection; its clean BLER, 0.0650, is close to that of LMMSE (0.0686).

Under attack it then has nothing protected to fall back on. The attacker degrades the least-squares estimate on which LMMSE rests, and the reference's virtual pass fraction falls from 0.60 to 0.13, 0.34 and 0.54 at 35, 75 and 150 m. The gates reject 0.920, 0.990 and 0.995 of the poisoned blocks, but the window BLER of Citadel-Rx, 0.779, 0.535 and 0.346, lies with those of online DeepSIC (0.661, 0.598, 0.374) and LMMSE (0.883, 0.581, 0.334): no receiver here is protected. As an excess over its clean twin (Table XIX), the BLER of Citadel-Rx rises more than that of online DeepSIC at every distance, with disjoint intervals at 35 m only: its clean BLER is the lower of the two (0.0650 against 0.169). The principle protects a memory that is worth protecting; when the network cannot track the channel the memory is not, and the receiver is left with the model-based reference, which the attacker reaches. At a user's maximum power the attacker at 35 and 75 m is also

TABLE XX
ONE-AT-A-TIME SENSITIVITY, WHITE-BOX A1 AT SIR 0 DB, TEST SEED 10: EXCESS SER WITH EACH SETTING MULTIPLIED BY 0.5, 0.75, 1.5 AND 2, AND THE LARGEST CLEAN FALSE REJECTION OVER THE FOUR. AT THE CHOSEN SETTINGS ($\times 1$) SEED 10 HAS AN EXCESS OF $+0.0130$ AND A CLEAN SER OF 0.0159, WITH 0/95 CLEAN BLOCKS FALSELY REJECTED.

Setting	$\times 0.5$	$\times 0.75$	$\times 1.5$	$\times 2$	Max false rej.
G1 slack $\tau_1 - 1$	$+0.0140$	$+0.0140$	$+0.0131$	$+0.0144$	0.000
G3 slack $\tau_3 - 1$	$+0.0134$	$+0.0131$	$+0.0143$	$+0.0130$	0.000
tightening $1-$floor	$+0.0133$	$+0.0135$	$+0.0135$	$+0.0137$	0.021
soft-step width	$+0.0129$	$+0.0132$	$+0.0141$	$+0.0146$	0.000
proximal weight λ	$+0.0142$	$+0.0130$	$+0.0133$	$+0.0139$	0.000
delta decay δ	$+0.0132$	$+0.0129$	$+0.0134$	$+0.0131$	0.000
CRC relative slack ρ	$+0.0130$	$+0.0138$	$+0.0138$	$+0.0133$	0.053
level-3 rate $\eta^{(3)}$	$+0.0136$	$+0.0141$	$+0.0141$	$+0.0130$	0.000

stronger than at the headline cell (0 dB): at the median drop its SIR is -26 dB at 35 m, below the -10 dB of the strongest attack in Table VII, and -3 dB at 75 m.

K. Sensitivity

Multiplying any one of eight settings by 0.5, 0.75, 1.5 or 2 (Table XX, seed 10) keeps the clean SER between 0.0156 and 0.0163 and the excess between $+0.0129$ and $+0.0146$ (0.0159 and $+0.0130$ at the chosen settings), with a clean false rejection of at most 0.053. That largest value, at half the CRC relative slack, and the one at twice the tightening exceed the 2% cap of the calibration rule (Section V); for the excess SER none of these eight settings is critical within a factor of two. Changing the CRC slack and the G1 slack together, by 0.5 or 2 each (the 4 corners), gives $+0.0134$ to $+0.0137$ at a false rejection of at most 0.053. The test uses one seed and four multipliers per setting, so it shows the absence of a cliff near the chosen values, not an optimum.

VII. DISCUSSION

What the results support. A test-time receiver can be protected against pilot poisoning without locating the attacker if what it keeps is checked against something the poisoned pilots of the block do not reach (in Citadel-Rx the slow memory, through the anchor detector, and the CRC of the data) and if the blocks it rejects are detected with that memory. Without this fallback the excess is $+0.0859$ instead of $+0.0199$ (seeds 10–14). CRC-Rollback, a CRC test at a fixed slack with rollback, judges against LMMSE on the poisoned pilots alone and fails, and so does Citadel-Rx with its own gates, slack and CRC selection but a single memory ($+0.316$). The failure comes from the reference, not from the slack: with LMMSE as the only reference, Citadel-Rx's own relative slack loosens as the attack degrades that reference. What remains under attack, an excess of $+0.0188$ at the headline cell, is the price of rejecting: in the forty attacked blocks the receiver detects with a memory that did not adapt to them, and the CRC selection loses most of its gain, because its LMMSE alternative uses the poisoned estimate (its pass fraction falls from 0.71 to 0.20); without the selection the excess is $+0.0053$, against $+0.0199$ on seeds 10–14. Under the 13 attack variants of Table IX at SIR 0 dB, those adapted to the defence, the physical attackers and the signal-referenced budget, all at full budget and rejected on nearly every attacked block, the excess stays between $+0.0171$ and $+0.0198$, not distinguishable from plain A1 with five seeds; the ordering holds for the 8 of them also run against online DeepSIC and, for plain A1, with shorter blocks, with eight users and 32 antennas and with a second network backbone. Over 2000 clean blocks (2 seeds) the SER of Citadel-Rx does not drift (0.0149 over blocks 5–499, 0.0149 over the last 500), and 1/3990 clean blocks are rejected. A slow perturbation kept small enough to be accepted is the case the retention bounds address: under A3* at 5% (1/600 attacked blocks rejected) the paired displacement of the slowest level reached 8.48 in the 300 attacked blocks, against a clean movement of 26.8 and a design bound of 90.0 (over 900 blocks 14.94, against 43.2 and 261.0), and cost $+0.0017$ in SER. The gain without attack is mostly the CRC selection, which any receiver with a CRC can use.

Where it fails. The defence is only as good as its references and its network. An attacker that also perturbs the data (A4) degrades the references, so that more poisoned blocks are accepted (only 152/200 rejected at $\alpha = 0.25$, 155/200 with 20 PGD steps), at an excess of up to $+0.0478$. The final abstain state brings the excess of A4 near that of A1 ($+0.0213$ and $+0.0249$ at $\alpha = 0.25$ and 0.75, against $+0.0192$, with overlapping intervals), but it abstains after some legitimate redraws, and its first version rejects every block of its clean runs. Where the network passes far fewer CRCs than the reference on clean blocks (0.11 against 0.93 at 16-QAM, a virtual 0.03 against 0.60 in the coded scene), the relative gates reject nearly every clean block and the receiver runs on its anchor and, through the CRC selection, on LMMSE where LMMSE passes. At 16-QAM the sub-blocks where nothing passes keep the anchor's decisions, so its clean SER (0.120) is worse than that of online DeepSIC (0.0512) and of LMMSE (0.0094), and under attack all three fail, Citadel-Rx with the smallest excess ($+0.558$ against $+0.932$ and $+0.951$). In the coded 3GPP scene, whose channels the networks do not track, it inherits LMMSE's exposure to poisoned pilots: its excess BLER ($+0.675$, $+0.432$ and $+0.243$ at 35, 75 and 150 m) exceeds that of undefended online DeepSIC ($+0.423$, $+0.360$, $+0.136$) at every distance, with disjoint intervals at 35 m only, and no receiver is protected. Strong poisoning during a legitimate change denies adaptation while it lasts, and its cost outlasts it (excess $+0.0543$ in the window, $+0.0511$ in the ten blocks after it); streak recovery gives $+0.0461$ and $+0.0462$ (5 seeds, no interval), so most of the cost remains. Without attack, the gates slow legitimate change: after a channel redraw they reject a fraction 0.180 of the clean blocks, and the SER over the remaining blocks is 0.250 against 0.224 for online DeepSIC; on the weaker second backbone they reject 103/475 clean blocks. Per-user gating does not isolate an attack on one user, because the block-level gates still reject the block. The gates do not address jamming, which at the same SIR harms every receiver more than poisoning harms Citadel-Rx.

Scope and limitations. The evaluation is link-level simulation only. The SED benchmark channel is real-valued with deterministic fades and identical for every seed, so the test seeds do not test a new channel (Section V). On this channel,

maximum-likelihood detection with the LS estimate has a lower clean SER ([p1.ml.wb0.clean?], Table VII) than every deep receiver, so the evaluation tests the protection of an adapting memory, not the case for deep receivers. The A4 surrogate holds its references at their values on the unattacked block; with the soft-gate objective, A3* does not steer toward a target model; the channel tracker and the surprise statistic are updated on rejected blocks too, so attacks on the gates' own state were not evaluated. Five seeds and marginal intervals support "not distinguishable", not equivalence. The coded experiment uses NR LDPC code blocks with scrambling over stochastic 3GPP urban-micro drops (not a ray-traced scene) with four antennas, one narrowband channel per slot, ideal power control and hard-decision decoding; the other experiments use CRC-8 sub-blocks over the SED model with cosine fading (COST2100 traces enter the calibration only). The headline cell has ten test seeds and most other experiments five; the slow small-budget attack, the first abstain version, the unstretched 4×16 runs and the long clean horizon have two and the sensitivity test one, so seed-level tests are possible only at the headline cell (with five seeds the smallest attainable two-sided p is 0.0625). The theory bounds the displacement of the slowest level, not the error rate; the design bound holds for any attacker but is coarse, and the structural bound needs the paired clean run, so it is a diagnostic of an experiment, not a run-time certificate. All results of the first version used the earlier consolidation rule, which Section III corrects, and are superseded by this evaluation, except the clean G1 statistics of Section IV, from clean runs, on which the two rules coincide unless the verdicts of a period sum to less than one, and the first Lipschitz probes of Remark 1, which concern only the fast learner, unchanged by the correction. Times were measured on CPUs with several runs in parallel and are complemented by multiply-accumulate counts; no GPU measurement is reported. At 3.03 s per block on one CPU core at the headline cell, the receiver is far from real time; a deployable version would start from the variants with 30 fast steps or with the last layer alone adaptable (Section VI-H), whose latency on an accelerator was not measured.

VIII. Conclusion

This paper argued that a test-time neural receiver under pilot poisoning should consolidate into persistent memory only what a label-free physical check verifies, and that the check should judge every update against that memory. Citadel-Rx realises this with gated, clipped consolidation into two slow levels, a fallback to the slow memory and per-sub-block CRC selection, and a clip that bounds how far any attacker can move the slowest level per consolidation. On held-out seeds of the stationary benchmark channel, white-box poisoning (A1) at SIR 0 dB raised its SER by $+0.0188$, against $+0.768$ for tuned online DeepSIC and $+0.271$ for CRC-Rollback, a CRC test with rollback judged against LMMSE alone, at a false-rejection rate of 0.001; after a legitimate channel redraw the false-rejection rate was 0.180 and the SER 0.251, against 0.224 for online DeepSIC. Citadel-Rx's own gates with a single memory gave $+0.316$ on seeds 10–14. The adapted attacks tested, all at full budget and rejected on nearly every attacked block, gave an excess not distinguishable from plain A1 with five seeds, and for plain A1 the ordering held on a second backbone and on larger arrays. An attack that also perturbs the data (A4: $+0.0337$ with 6 PGD steps and $+0.0478$ with 20, the worst attack found without a channel change) and poisoning during a legitimate channel redraw (A1 at SIR 0 dB: $+0.0543$ in the window and $+0.0511$ after it; the channel-estimate spoof at 10 dB: $+0.0968$) raised the excess. A slow small-budget attack that the gates accepted (A3* at 5%, 1/600 blocks rejected) cost $+0.0017$ in SER, and the paired displacement of the slowest level reached 8.48 in the window, against a clean movement of 26.8 and a design bound of 90.0. The open problems are the ones the evaluation exposed. Three concern the gates: references an attacker cannot degrade through the data, adaptation to a legitimate change while poisoned blocks are rejected, and an abstain statistic that survives legitimate change (a per-user verified residual is the next candidate). Two concern the network: networks that track the channels of the coded 3GPP scene, without which that link is not protected (there the receiver already runs on its anchor and on LMMSE, which the attacker reaches), and a fallback for networks weaker than the model-based detector, which the clean 16-QAM results call for.

Acknowledgment

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References

[1] T. O'Shea and J. Hoydis, "An introduction to deep learning for the physical layer," *IEEE Trans. Cogn. Commun. Netw.*, vol. 3, no. 4, pp. 563–575, Dec. 2017.

[2] M. Honkala, D. Korpi, and J. M. J. Huttunen, "DeepRx: Fully convolutional deep learning receiver," *IEEE Trans. Wireless Commun.*, vol. 20, no. 6, pp. 3925–3940, Jun. 2021.

[3] N. Samuel, T. Diskin, and A. Wiesel, "Learning to detect," *IEEE Trans. Signal Process.*, vol. 67, no. 10, pp. 2554–2564, May 2019.

[4] N. Shlezinger, J. Whang, Y. C. Eldar, and A. G. Dimakis, "Model-based deep learning," *Proc. IEEE*, vol. 111, no. 5, pp. 465–499, May 2023.

[5] N. Shlezinger, R. Fu, and Y. C. Eldar, "DeepSIC: Deep soft interference cancellation for multiuser MIMO detection," *IEEE Trans. Wireless Commun.*, vol. 20, no. 2, pp. 1349–1362, Feb. 2021.

[6] T. Raviv, S. Park, O. Simeone, Y. C. Eldar, and N. Shlezinger, "Online meta-learning for hybrid model-based deep receivers," *IEEE Trans. Wireless Commun.*, vol. 22, no. 10, pp. 6415–6431, Oct. 2023.

[7] K. Wu, W. Jiang, D. Niyato, Y. Li, and C. Luo, "Transfer-based adversarial poisoning attacks for online (MIMO-)deep receviers," arXiv:2409.02430, Sep. 2024.

[8] D. Kapetanović, G. Zheng, K.-K. Wong, and B. Ottersten, "Detection of pilot contamination attack using random training and massive MIMO," in *Proc. IEEE 24th Annu. Int. Symp. Pers., Indoor, Mobile Radio Commun. (PIMRC)*, 2013, pp. 13–18.

[9] J. K. Tugnait, "Self-contamination for detection of pilot contamination attack in multiple antenna systems," *IEEE Wireless Commun. Lett.*, vol. 4, no. 5, pp. 525–528, Oct. 2015.

[10] N. Shlezinger, N. Farsad, Y. C. Eldar, and A. J. Goldsmith, "ViterbiNet: A deep learning based Viterbi algorithm for symbol detection," *IEEE Trans. Wireless Commun.*, vol. 19, no. 5, pp. 3319–3331, May 2020.

[11] S. Schibisch, S. Cammerer, S. Dörner, J. Hoydis, and S. ten Brink, "Online label recovery for deep learning-based communication through error correcting codes," in *Proc. 15th Int. Symp. Wireless Commun. Syst. (ISWCS)*, 2018, pp. 1–5.

[12] S. Niu, J. Wu, Y. Zhang, Y. Chen, S. Zheng, P. Zhao, and M. Tan, "Efficient test-time model adaptation without forgetting," in *Proc. 39th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 162, 2022, pp. 16888–16905.

[13] Q. Wang, O. Fink, L. Van Gool, and D. Dai, "Continual test-time domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 7201–7211.

[14] S. Niu, J. Wu, Y. Zhang, Z. Wen, Y. Chen, P. Zhao, and M. Tan, "Towards stable test-time adaptation in dynamic wild world," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.

[15] L. Yuan, B. Xie, and S. Li, "Robust test-time adaptation in dynamic scenarios," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 15922–15932.

[16] H. Park, J. Hwang, S. Mun, S. Park, and J. Ok, "MedBN: Robust test-time adaptation against malicious test samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 5997–6007.

[17] T. Wu, F. Jia, X. Qi, J. T. Wang, V. Sehwag, S. Mahloujifar, and P. Mittal, "Uncovering adversarial risks of test-time adaptation," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 202, 2023, pp. 37456–37495.

[18] X. Li, Y. Grandvalet, and F. Davoine, "Explicit inductive bias for transfer learning with convolutional networks," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 80, 2018, pp. 2825–2834.

[19] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, and R. Hadsell, "Overcoming catastrophic forgetting in neural networks," *Proc. Natl. Acad. Sci. USA*, vol. 114, no. 13, pp. 3521–3526, Mar. 2017.

[20] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 1195–1204.

[21] A. Behrouz, M. Razaviyayn, P. Zhong, and V. Mirrokni, "Nested learning: The illusion of deep learning architectures," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025; arXiv:2512.24695.

[22] A. Behrouz, P. Zhong, and V. Mirrokni, "Titans: Learning to memorize at test time," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025; arXiv:2501.00663.

[23] X. Zhou, B. Maham, and A. Hjørungnes, "Pilot contamination for active eavesdropping," *IEEE Trans. Wireless Commun.*, vol. 11, no. 3, pp. 903–907, Mar. 2012.

[24] D. Kapetanović, G. Zheng, and F. Rusek, "Physical layer security for massive MIMO: An overview on passive eavesdropping and active attacks," *IEEE Commun. Mag.*, vol. 53, no. 6, pp. 21–27, Jun. 2015.

[25] N. Uzlaner, T. Raviv, N. Shlezinger, and K. Todros, "Asynchronous online adaptation via modular drift detection for deep receivers," arXiv:2407.09134, Jul. 2024.

[26] Y. Gusakov, O. Simeone, T. Routtenberg, and N. Shlezinger, "Online learning of modular Bayesian deep receivers: Single-step adaptation with streaming data," arXiv:2511.06045, Nov. 2025.

[27] T.-H. Hoang, D. M. Vo, and M. N. Do, "Persistent test-time adaptation in recurring testing scenarios," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 37, 2024.

[28] T. Cong, X. He, Y. Shen, and Y. Zhang, "Test-time poisoning attacks against test-time adaptation models," in *Proc. IEEE Symp. Secur. Privacy (SP)*, 2024, pp. 1306–1324.

[29] Y. Su, Y. Li, N. Liu, K. Jia, X. Yang, C.-S. Foo, and X. Xu, "On the adversarial risk of test time adaptation: An investigation into realistic test-time data poisoning," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.

[30] O. Press, S. Schneider, M. Kümmerer, and M. Bethge, "RDumb: A simple approach that questions our progress in continual test-time adaptation," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023.

[31] P. Blanchard, E. M. El Mhamdi, R. Guerraoui, and J. Stainer, "Machine learning with adversaries: Byzantine tolerant gradient descent," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 119–129.

[32] J. Steinhardt, P. W. Koh, and P. Liang, "Certified defenses for data poisoning attacks," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 3517–3529.

[33] Y. Hu and L. Duan, "Theory of continual learning against data poisoning attacks," in *Proc. 43rd Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 306, 2026, pp. 44906–44935.

[34] J. L. McClelland, B. L. McNaughton, and R. C. O'Reilly, "Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory," *Psychol. Rev.*, vol. 102, no. 3, pp. 419–457, Jul. 1995.

[35] D. Kumaran, D. Hassabis, and J. L. McClelland, "What learning systems do intelligent agents need? Complementary learning systems theory updated," *Trends Cogn. Sci.*, vol. 20, no. 7, pp. 512–534, Jul. 2016.

[36] M. Sadeghi and E. G. Larsson, "Adversarial attacks on deep-learning based radio signal classification," *IEEE Wireless Commun. Lett.*, vol. 8, no. 1, pp. 213–216, Feb. 2019.

[37] M. Sadeghi and E. G. Larsson, "Physical adversarial attacks against end-to-end autoencoder communication systems," *IEEE Commun. Lett.*, vol. 23, no. 5, pp. 847–850, May 2019.

[38] Y. E. Sagduyu, Y. Shi, and T. Erpek, "Adversarial deep learning for over-the-air spectrum poisoning attacks," *IEEE Trans. Mobile Comput.*, vol. 20, no. 2, pp. 306–319, 2021.

[39] A. Athalye, N. Carlini, and D. Wagner, "Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 80, 2018, pp. 274–283.

[40] N. Carlini, A. Athalye, N. Papernot, W. Brendel, J. Rauber, D. Tsipras, I. Goodfellow, A. Madry, and A. Kurakin, "On evaluating adversarial robustness," arXiv:1902.06705, 2019.

[41] F. Tramèr, N. Carlini, W. Brendel, and A. Madry, "On adaptive attacks to adversarial example defenses," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 1633–1645.

[42] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 70, 2017, pp. 1126–1135.

[43] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," TR 38.901, V19.5.0, 2026, Table 7.4.1-1.

[44] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018.

[45] A. Athalye, L. Engstrom, A. Ilyas, and K. Kwok, "Synthesizing robust adversarial examples," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 80, 2018, pp. 284–293.

[46] D. Wierstra, T. Schaul, T. Glasmachers, Y. Sun, J. Peters, and J. Schmidhuber, "Natural evolution strategies," *J. Mach. Learn. Res.*, vol. 15, no. 27, pp. 949–980, 2014.

[47] A. Ilyas, L. Engstrom, A. Athalye, and J. Lin, "Black-box adversarial attacks with limited queries and information," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 80, 2018, pp. 2137–2146.

[48] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *J. Amer. Statist. Assoc.*, vol. 22, no. 158, pp. 209–212, Jun. 1927.

[49] T. Raviv, S. Park, O. Simeone, Y. C. Eldar, and N. Shlezinger, "Adaptive and flexible model-based AI for deep receivers in dynamic channels," *IEEE Wireless Commun.*, vol. 31, no. 4, pp. 163–169, 2024.

[50] W. Hoeffding, "Probability inequalities for sums of bounded random variables," *J. Amer. Statist. Assoc.*, vol. 58, no. 301, pp. 13–30, Mar. 1963.

[51] P. J. Huber, "Robust estimation of a location parameter," *Ann. Math. Statist.*, vol. 35, no. 1, pp. 73–101, Mar. 1964.

[52] 3GPP, "Evolved Universal Terrestrial Radio Access (E-UTRA); Multiplexing and channel coding," TS 36.212, V19.1.0, 2025, Sec. 5.1.1.

[53] L. Liu, C. Oestges, J. Poutanen, K. Haneda, P. Vainikainen, F. Quitin, F. Tufvesson, and P. De Doncker, "The COST 2100 MIMO channel model," *IEEE Wireless Commun.*, vol. 19, no. 6, pp. 92–99, Dec. 2012.

[54] 3GPP, "NR; Multiplexing and channel coding," TS 38.212, V19.2.0, 2025, Secs. 5.1 and 5.3.2.

[55] 3GPP, "NR; Physical channels and modulation," TS 38.211, V19.2.0 (ETSI TS 138 211 V19.2.0, Feb. 2026), Secs. 5.2.1 and 6.3.1.1.

[56] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *Ann. Statist.*, vol. 17, no. 3, pp. 1217–1241, Sep. 1989.

[57] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bull.*, vol. 1, no. 6, pp. 80–83, Dec. 1945.

[58] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.

[59] J. Hoydis, S. Cammerer, F. Ait Aoudia, M. Nimier-David, L. Maggi, G. Marcus, A. Vem, and A. Keller, "Sionna," software, 2022; version 2.2.0, Sep. 2026. [Online]. Available: https://nvlabs.github.io/sionna/